

Quantized Supertwistors, (Split-) Octonions, and Unification

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We explore twistor geometry and show that it is possible to obtain the superconformal, octonionic, and split-octonionic algebras from quantized $\mathcal{N} = 4$ supertwistors. We motivate a Unified theory with internal symmetries from octonions and spacetime symmetry from split-octonions respectively and review some of the work done along this line.

Statement of original research

Chapter 1 of this essay contains an introductory overview; Chapters 2 and 3 contain a literature review of Twistor theory; Chapter 4 is based on the work done in Ref. [1] and also describes original work, whose results are summarized in Chapter 4 and Appendices D and E; Chapter 5 reproduces the results in Ref. [2].

1 Introduction

The Standard Model is experimentally the closest verified model we have for all of nature's forces and matter particles. The last 40 years of collider physics seem to suggest that it is unlikely to discover new particles at low energies. Therefore, we need to take the Standard Model seriously. How-

ever, the Standard Model is experimentally determined with 26 unexplained dimensionless parameters. The presence of these parameters makes it debatable whether the Standard Model is a fundamental theory of nature or just a modeling of parameters that explains experiments. After a long time, the dust is settling and we need an underlying math that theoretically justifies the Standard Model. Despite its success, the Standard Model does not explain many aspects of nature including dark matter, matter/anti-matter asymmetry, neutrino masses and oscillation, dimensionless coupling constants, the cosmological constant, and a renormalizable theory of quantum gravity. It is hoped that theories that predict a minimalistic variation of the Standard Model should explain these phenomena. With this hope, we explore twistors and octonions in order to come up with an underlying mathematics behind the Standard Model that not only explains its uniqueness but also tries to answer questions beyond the Standard Model.

There are only four finite-dimensional normed division algebras over $\mathbb{R}$. The real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$, the quaternions $\mathbb{H}$, and the octonions $\mathbb{O}$. Normed division algebras have played a significant role in mathematics, from introducing the notion of associativity to the development of vector analysis [3], the discovery of Clifford algebras [4], and the classification of classical and exceptional compact semi-simple Lie algebras [5]. Real numbers and complex numbers have an undebatable role to play in physics. The other two normed division algebras however have not been used in their full glory by physicists. Quaternions were invented by Hamilton in 1843 to explain rotations in 3D. Quaternions generate the Lie group $SU(2)$ and are therefore useful in explaining rotations and spin in quantum mechanics. Quaternionic matrices also generate symplectic groups, that explain the symmetry in symplectomorphic phase spaces. The first application of octonions in physics was in 1934 by Jordan, Von Neumann, and Wigner in their work on algebraic quantum mechanics [6]. In the 1980s it was realized that 2×2 special linear matrices with entries in $\mathbb{R}, \mathbb{C}, \mathbb{H}$, and $\mathbb{O}$ respectively are the spinor groups for 3, 4, 6, and 10 dimensional Minkowski spacetimes that are relevant for supersymmetric theories [7]. It was only after the discovery of quarks and the emergence of the Standard Model that people started looking into the significance of octonions in physics. Gunaydin and Gursay (1974) were the first to talk about quark statistics and octonions [8]. Dixon in 1994 wrote a book on the algebraic design of physics talking about the role of octonions in the Standard Model [9]. Furey [10] and Dixon [9] construct $Cl(6)$ spinors via Witt decomposition (as reviewed by Ablamowicz in 1995 [11]) of

octonions to show that these spinors could be the one generation of Standard Model fermions. Starting from the 1980s people also started looking for Grand Unified Theories using exceptional Jordan algebra, Exceptional Lie groups, and octonions [2, 10, 12–17]. The Freudenthal-Tits Magic Square provided a unified description of compact semi-simple classical and exceptional Lie algebras [18–20]. The triality decomposition of the magic square as proposed by Ramond in 1977 [21] is being used by many authors to explain the three generations of fermions in the Standard Model [12]. For a thorough review of octonions, one should refer to [22].

Concurrently in the 1960s, Penrose and his collaborators developed the twistor theory [23]. The original motivation for twistor theory was to describe spacetime as an emergent object from a more fundamental underlying spinor geometry [24, 25]. Twistors describe spacetime non-locally by considering null rays to be more fundamental objects than spacetime points. Penrose argues that quantizing spacetime with this description keeps the causal structure intact which is not the case if spacetime points are treated to be more fundamental [26]. Work in twistor theory led to a lot of progress in spinor geometry and the Penrose-Ward transform provided a way to look at massless quantum fields as integrals of holomorphic homogenous functions on the twistor space [27]. Ever since Witten’s paper on twistor string correspondence in 2003, [28], people began taking an interest in twistor theory to calculate scattering amplitudes. A slightly less-known field of research that started in the ’70s by Penrose and his collaborators is the field of twistor particle physics [29, 30]. Twistor space is a complexified space, taking different real slices of the twistor space results in spacetimes with different signatures. However, some argue that not all of twistor theory is about spacetimes, there exist complex slices in twistor space that do not correspond to real spacetimes but to the space of the internal symmetry groups of the Standard Model. Recent work by Woit is motivated along these lines [31].

How do twistors and octonions connect together? And how do both of them connect to the Standard Model? Work by Hitchin in the past has shown that twistor spaces corresponding to 10 and 6 dimensions can be explained using quaternionic and octonionic spinors respectively [32]. However, very recent work by Penrose shows that octonions are related to 4-dimensional twistors too [1]. Penrose shows that using quantized Lorentzian twistors as the basis we can construct the algebra of split-octonions. In this project, we further show that one can obtain both octonionic and split-octonionic algebra if one applies Penrose’s procedure to quantized $\mathcal{N} = 4$ supertwistors. Recent

work by Krasnov [2] and Dray, Manogue [16] shows that split-octonions and octonions can be used together to obtain the standard model gauge group along with Lorentz symmetry. We review the $\text{Spin}(11, 3)$ unification model based on octonions and split-octonions [2] and motivate its relation to quantized supertwistors.

2 Mathematical Preliminaries

In this section, we briefly present the mathematical requisites for this project. We briefly introduce (split-)division algebras followed by 2-spinor notation which is useful for twistor theory. In the last two subsections, we explain twistor theory and how to obtain real spacetimes from twistor space.

2.1 Division and Composition Algebras

A composition algebra $\mathcal{A}$ is an algebra over a field $\mathcal{F}$ equipped with a non-degenerate quadratic form $|\cdot|^2 : \mathcal{A} \rightarrow \mathcal{F}$ such that:

$$|xy|^2 = |x|^2|y|^2. \quad (1)$$

Any $x \in \mathcal{A}$ can be written as $x = \text{Re}(x) + \text{Im}(x)$, where $\text{Re}(x) \in \mathbb{R}$ but $\text{Im}(x) \notin \mathbb{R}$. The conjugation operation can be defined in a way such that $\bar{x} = \text{Re}(x) - \text{Im}(x)$. Then a quadratic form is naturally given by $x\bar{x} = \bar{x}x$.

All composition algebras are alternative, i.e., if we define the associator as:

$$[x, y, z] = x(yz) - (xy)z, \quad (2)$$

then for alternative algebras, the associator is completely anti-symmetric in its arguments:

$$[x, x, z] = [x, z, x] = [z, x, x] = 0. \quad (3)$$

If the quadratic form $|x|^2$ is positive definite then the composition algebra is a division algebra. A division algebra can be loosely defined as an algebra where the notion of divisibility exists; moreover, if the inner product between two elements of the algebra is 0 then at least one of the two elements is 0.

$$xy = 0 \Rightarrow x = 0 \quad \text{or} \quad y = 0 \quad (4)$$

According to Hurwitz's theorem [33], there are only four normed division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}$, and $\mathbb{O}$ over $\mathbb{R}$. And beyond this, there are only three more composition algebras over $\mathbb{R}$, namely $\tilde{\mathbb{C}}, \tilde{\mathbb{H}}, \tilde{\mathbb{O}}$ (the split versions of $\mathbb{R}, \mathbb{C}, \mathbb{H}$, and $\mathbb{O}$). These can be constructed using the Cayley-Dickson method [34].

The quaternions $\mathbb{H}$ comprise one real unit and three imaginary units i, j, k . The quaternions have the following multiplication rule:

$$i^2 = j^2 = k^2 = ijk = -1 \quad (5)$$

$$ij + ji = jk + kj = ki + ik = 0 \quad (6)$$

$$ij = k, jk = i, ki = j. \quad (7)$$

Note that this is the algebra of the Pauli matrices multiplied with 'i'.

The octonions have one real unit and 7 imaginary units $e_i^2 = -1$ for $i = 1, 2, \dots, 7$. Just like the quaternions, the imaginary units of octonions anti-commute with each other. The multiplication of octonions can be understood using the projective plane $\mathbb{OP}^1$, called the Fano plane, see figure 1.

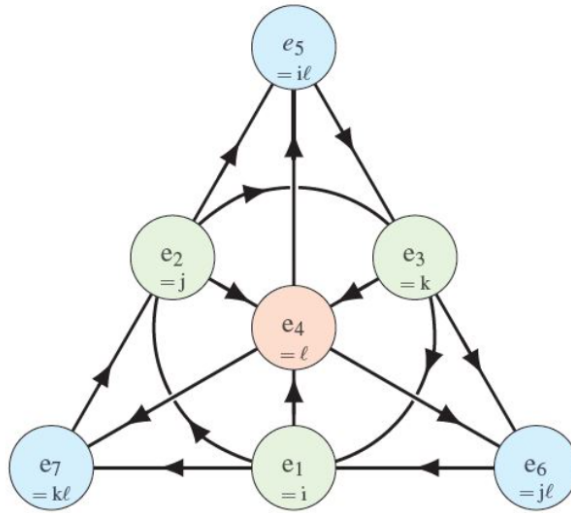


Figure 1: The Fano Plane, credits: Ncatlab

There are seven quaternionic subsets in the Fano plane, given by the three sides of the triangle, the three altitudes, and the incircle (Fig.1). Following the arrows will give us a positive sign whereas going against the arrows will

give us a negative sign. For example, let's check the multiplication rule along one of the quaternionic subsets given by the incircle of figure 1, $e_1e_2 = e_3$, $e_2e_3 = e_1$, $e_1e_3 = -e_2$.

A general octonionic number can be written as:

$$\omega = a_0 + a_1e_1 + a_2e_2 + a_3e_3 + a_4e_4 + a_5e_5 + a_6e_6 + a_7e_7 \in \mathbb{O}, \quad a_i \in \mathbb{R}. \quad (8)$$

Just as with complex numbers, octonionic conjugate and the quadratic form are given as follows:

$$\bar{\omega} = a_0 - a_1e_1 - a_2e_2 - a_3e_3 - a_4e_4 - a_5e_5 - a_6e_6 - a_7e_7 \quad (9a)$$

$$|\omega|^2 = \omega\bar{\omega} = \bar{\omega}\omega = a_0^2 + a_1^2 + a_2^2 + a_3^2 + a_4^2 + a_5^2 + a_6^2 + a_7^2. \quad (9b)$$

Division algebras have a positive definite quadratic form but we can construct composition algebras using the Cayley-Dickson method that have a split signature quadratic form. Such algebras are called split-division algebras and we will denote them as $\tilde{\mathbb{R}} \cong \mathbb{R}$, $\tilde{\mathbb{C}}$, $\tilde{\mathbb{H}}$, and $\tilde{\mathbb{O}}$. In this essay, we are particularly interested in split-octonions that can be written as:

$$\omega = a_0 + a_1\tilde{e}_1 + a_2\tilde{e}_2 + a_3\tilde{e}_3 + a_4\tilde{e}_4 + a_5\tilde{e}_5 + a_6\tilde{e}_6 + a_7\tilde{e}_7 \in \tilde{\mathbb{O}}. \quad (10)$$

The multiplication rule for split-octonions can again be given by the Fano plane (Fig. 1) but note the following difference:

$$\tilde{e}_1^2 = \tilde{e}_2^2 = \tilde{e}_3^2 = -1, \quad \tilde{e}_4^2 = \tilde{e}_5^2 = \tilde{e}_6^2 = \tilde{e}_7^2 = 1. \quad (11)$$

Because of this difference, the split-octonionic quadratic form is given by:

$$|\omega|^2 = a_0^2 + a_1^2 + a_2^2 + a_3^2 - a_4^2 - a_5^2 - a_6^2 - a_7^2. \quad (12)$$

This (4, 4) split-signature justifies the name split-octonions. Note that $\tilde{\mathbb{O}}$ is not a division algebra.

The (split-)division algebras have a very important role to play in the categorification of semi-simple Lie algebras and Jordan algebras. Some details are in Appendix A, for further details, readers are referred to [22].

2.2 2-Spinor Notation

The isometry group of spacetime is the Lorentz group $\text{SO}(1, 3)$. The double cover of $\text{SO}(1, 3)$ is $\text{SL}(2, \mathbb{C})$ and can be written as $\text{SU}(2)_L \times i\text{SU}(2)_R$.

An object that transforms under $SU(2)_L$ but is invariant under the $SU(2)_R$ representation of $SL(2, \mathbb{C})$ is called a left-handed spinor and is represented as $(\frac{1}{2}, 0)$. Similarly, right-handed spinors transform under $SU(2)_R$ but not under $SU(2)_L$ and are represented as $(0, \frac{1}{2})$. A Dirac spinor is a direct sum of these two spinors and can be written as $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$. A spacetime vector transforms under both copies of $SU(2)$ and is represented as $(\frac{1}{2}, \frac{1}{2})$.

In the 2-spinor notation, a left-handed spinor is written as x^A where A runs from 0 to 1. A right-handed spinor is written as $x^{\dot{A}}$ with a dotted index, again $\dot{A}$ runs from 0 to 1. A spacetime vector V is written with both dotted and undotted indices as $V^{A\dot{A}}$. The following equation expresses a vector V with spacetime components $V^a = (V^0, V^1, V^2, V^3)$ in the 2-spinor notation:

$$V^{A\dot{A}} = \frac{1}{\sqrt{2}} V^a \sigma_a^{A\dot{A}}. \quad (13)$$

Here σ_a are the Pauli matrices with $\sigma_0 = I$. Therefore:

$$V^{A\dot{A}} = \frac{1}{\sqrt{2}} \begin{pmatrix} V_0 + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & V_0 - V_3 \end{pmatrix}. \quad (14)$$

Similarly, any tensor with vector indices can be upgraded to a tensor with spinor indices.

$$T^{ab} \rightarrow T^{A\dot{A}B\dot{B}} \quad (15)$$

Throughout this essay, we will use small Latin letters to denote spacetime indices, capital Latin letters to denote spinor indices, and Greek letters to denote Twistor indices which will be introduced in section 2.3.

Note that the Lorentz invariant norm of this vector is given as:

$$\langle V^a \rangle = V_0^2 - V_1^2 - V_2^2 - V_3^2 = 2\text{Det}(V^{A\dot{A}}). \quad (16)$$

Using the expression for determinant, we have:

$$2\text{Det}(V^{A\dot{A}}) = \epsilon_{AB}\epsilon_{\dot{A}\dot{B}} V^{A\dot{A}} V^{B\dot{B}} = \eta_{ab} V^a V^b, \quad (17)$$

where ϵ is the 2D Levi-Civita matrix:

$$\epsilon_{AB} = \epsilon_{\dot{A}\dot{B}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (18)$$

Therefore, using 17, we establish the following equivalence:

$$\eta_{ab} \rightarrow \epsilon_{AB}\epsilon_{\dot{A}\dot{B}}. \quad (19)$$

Expressing $\epsilon_{AB}\epsilon_{\dot{A}\dot{B}}V^{A\dot{A}}$ as $V_{B\dot{B}}$ we note how the Levi-Civita matrix acts on the spinor indices:

$$\omega^A\epsilon_{AB} = \omega_A, \quad \epsilon^{AB}\pi_B = \pi^A. \quad (20)$$

It is important to keep track of the order of spinor indices since the ϵ matrix is skew-symmetric. A general rule of thumb is that ϵ_{AB} lowers the repeated index ‘A’ on the right while ϵ^{AB} raises the repeated index ‘B’ on the left.

Note the following relation:

$$\Lambda_c{}^a\eta_{ab}\Lambda_d{}^b = \eta_{cd}. \quad (21)$$

In spinor indices, the Lorentz transformation matrix can be written as:

$$\Lambda_{ab} \rightarrow l_{AB}\bar{l}_{\dot{A}\dot{B}}. \quad (22)$$

Here l_{AB} and $\bar{l}_{\dot{A}\dot{B}}$ are elements of $\text{SL}(2, \mathbb{C})$. The identity in eqn. (21) can be written as the preservation of the Levi-Civita matrix under Lorentz action:

$$l_A{}^Cl_B{}^D\epsilon_{CD} = \epsilon_{AB}. \quad (23)$$

A useful property that can be checked by the reader is that any null vector can be written in the following way:

$$V^a := V^{A\dot{A}} = \pi^A\bar{\pi}^{\dot{A}}. \quad (24)$$

It is important to emphasize that the vector V presented in eqn. (13, 14) lives in Minkowski spacetime $\mathcal{M}$, but in general, we can complexify the spacetime indices V^a so that V now lives in a complexified spacetime $\mathcal{M}^{\mathbb{C}}$ (V_0, V_1, V_2, V_3 in eqn. (14) $\in \mathbb{C}$).

In [24], Penrose talks about the significance of this notation in emphasizing that spinors are the fundamental building blocks of vector spaces and therefore spinor indices should be used over vector indices to construct tensors. However, this notation provides no radical perspective change in understanding spacetime itself. In order to do so Penrose introduces the twistor space, a more fundamental space from which spacetime emerges.

2.3 Twistor Geometry

Twistor space is a four-dimensional complex space $\mathbb{T} = \mathbb{C}^4$. Complexified and compactified spacetime points are identified with points on the Grassmannian; $m \in G(2, \mathbb{C}^4) \cong \mathcal{M}^{\mathbb{C}}$. Let us define the tautological spinor bundle

over the spacetime as $\mathbb{S}$. The fibers S_m of the tautological spinor bundle are 2 complex dimensional and fix the spinor structure at each spacetime point m .

A point in the Grassmannian can be written as:

$$\omega_{ij} = e_i \otimes e_j - e_j \otimes e_i \in \Lambda^2(\mathbb{C}^4), \quad (25)$$

where e_i and e_j are any 2 vectors spanning the 2-dimensional subspace, $\dim_{\mathbb{C}} \Lambda^2(\mathbb{C}^4) = 6$. If we consider the projective space then $\mathbb{P}(\Lambda^2(\mathbb{C}^4)) = \mathbb{CP}^5$. We can further restrict ω_{ij} to lie on a quadric $\mathbb{Q}$ in $\mathbb{CP}^5$ constrained in the following way:

$$\omega_{12}\omega_{34} + \omega_{13}\omega_{42} + \omega_{14}\omega_{23} = 0. \quad (26)$$

This quadratic surface in $\mathbb{CP}^5$ is called the Klein quadric $\mathbb{Q}$ and it invokes the Klein-correspondence between points in $\mathbb{CP}^5$ and lines in $\mathbb{CP}^3$. In physics, this is the conformal compactification of spacetime that corresponds to taking the intersection of a lightlike (in order to obtain flat spacetime) hypersurface in $(p+1, q+1)$ spacetime with its null cone to obtain (p, q) dimensional spacetime with the conformal group $\text{SO}(p+1, q+1)$.

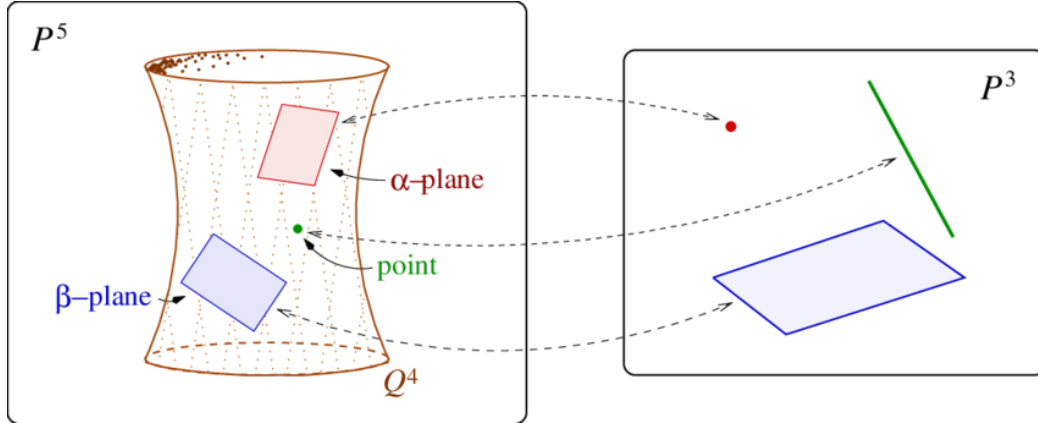


Figure 2: Pictorial description of Klein Correspondance. Points on the quadric $\mathbb{Q}$ in $\mathbb{CP}^5$ correspond to lines in $\mathbb{CP}^3$. Some planes on the quadric (α -planes) correspond to points in $\mathbb{CP}^3$ whereas some planes (β -planes) correspond to planes in $\mathbb{CP}^3$. Image Credits: [35]

Note that the projective twistor space $\mathbb{PT}$ is $\mathbb{CP}^3$. Using the Klein correspondence, we have seen that spacetime points correspond to lines in $\mathbb{CP}^3$

(projective twistor space). Lines in $\mathbb{PT}$ are given by the subspace $\mathbb{CP}^1$. Therefore spacetime points are parametrized by $\mathbb{CP}^1$ lines in twistor space. The complexified Lorentz algebra is $\mathfrak{spin}(4, \mathbb{C}) = \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$. A right-handed Weyl spinor $\pi_{\dot{A}} \in \mathbb{CP}^1$ of this algebra is a section of the tautological spinor bundle over the spacetime point $x^{A\dot{A}}$. The resulting point in the tautological spinor bundle is $(x^{A\dot{A}}, \pi_{\dot{A}}) \in \mathbb{PS}$. A point in the projective twistor space is $(ix^{A\dot{A}}\pi_{\dot{A}}, \pi_{\dot{A}}) \in \mathbb{PT}$ and can be obtained via a double fibration from the spinor bundle (see Fig. 3).

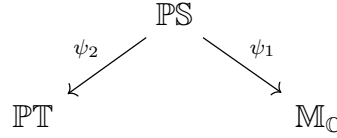


Figure 3: Double fibration from Spinor bundle to complex spacetime and Twistor space.

In figure 3, ψ_1 is the tautological fiber that takes $(x^{A\dot{A}}, \pi_{\dot{A}}) \in \mathbb{PS}$ to $x^{A\dot{A}} \in \mathcal{M}^{\mathbb{C}}$. The ψ_2 fibers take $(x^{A\dot{A}}, \pi_{\dot{A}}) \in \mathbb{PS}$ to a point in twistor space $(ix^{A\dot{A}}\pi_{\dot{A}}, \pi_{\dot{A}}) \in \mathbb{PT}$.

Therefore a general point in the twistor space $Z^\alpha \in \mathbb{CP}^3$ with $\alpha = 0, 1, 2, 3$ can be written as $(\omega^A, \pi_{\dot{A}})$ with $A, \dot{A} = 0, 1$. ω^A and $\pi_{\dot{A}}$ satisfy the following twistor incidence relation:

$$\omega^A = ix^{A\dot{A}}\pi_{\dot{A}}. \quad (27)$$

The conformal group of complexified spacetime is $\mathrm{SO}(6, \mathbb{C})$ with the double cover $\mathrm{SL}(4, \mathbb{C})$. $\mathrm{SL}(4, \mathbb{C})$ acts transitively on $\mathbb{T} = \mathbb{C}^4$ and therefore on its subspace $\mathbb{PT}$ and $G(2, \mathbb{C}^4)$. A twistor transforms as a spinor of the conformal group of complexified spacetime (**4** under $\mathrm{SL}(4, \mathbb{C})$). Let's see the action of $\Lambda \in \mathrm{SL}(4, \mathbb{C})$ on twistors.

$$\Lambda = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathrm{SL}(4, \mathbb{C})$$

Here Λ is a 4×4 linear matrix with determinant 1. A, B, C, D are 2×2 matrices.

$$\Lambda Z = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} \omega \\ \pi \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} ix\pi \\ \pi \end{pmatrix} \quad (28)$$

$$\Lambda Z = \begin{pmatrix} (Aix + B)\pi \\ (Cix + D)\pi \end{pmatrix} \quad (29a)$$

$$\Lambda Z = \begin{pmatrix} (Aix + B)(Cix + D)^{-1}(Cix + D)\pi \\ (Cix + D)\pi \end{pmatrix} \quad (29b)$$

$$\Lambda Z = Z' = \begin{pmatrix} ix'\pi' \\ \pi' \end{pmatrix} \quad (29c)$$

Comparing eqn. (29b) and (29c) we note how the spacetime point transforms under $SL(4, \mathbb{C})$:

$$ix \rightarrow (Aix + B)(Cix + D)^{-1}. \quad (30)$$

This is precisely the Möbius transformation that brings about conformal transformations in spacetime points. This takes the form of Lorentz transformation for $B = C = 0$:

$$x \rightarrow Ax D^{-1}. \quad (31)$$

The matrix Λ , in this case, is reduced to be in $SL(2, \mathbb{C}) \times SL(2, \mathbb{C})$ which is the double cover of the complexified Lorentz group $SO(4, \mathbb{C})$.

Twistors are physically very well motivated. Any point in twistor space corresponds to a null ray in Minkowski space. The incidence relation holds true for all spacetime points translated along the null ray $l^a = \tilde{\pi}^A \pi^{\dot{A}}$.

$$x^{A\dot{A}} \rightarrow x^{A\dot{A}} + \tilde{\pi}^A \pi^{\dot{A}} \quad (32)$$

$$\Rightarrow \omega^A = i(x^{A\dot{A}} + \tilde{\pi}^A \pi^{\dot{A}})\pi_{\dot{A}} \quad (33)$$

$$\Rightarrow \omega^A = i(x^{A\dot{A}})\pi_{\dot{A}} \quad (34)$$

The last equation follows from the fact that $\pi^{\dot{A}}\pi_{\dot{A}} = 0$. Therefore, the point $(\omega^A, \pi_{\dot{A}})$ satisfies the incidence relation for a family of spacetime points lying on the null ray $\tilde{\pi}^A \pi^{\dot{A}}$ parametrized by a complex number. If we impose reality condition on spacetime and restrict it to have Minkowski signature then the twistor $(\omega^A, \pi_{\dot{A}})$ uniquely describes the light ray $\tilde{\pi}^A \pi^{\dot{A}}$.

Thus we note that points in twistor space correspond to null rays in Minkowski space whereas $\mathbb{CP}^1$ lines in twistor space correspond to spacetime points in Minkowski space.

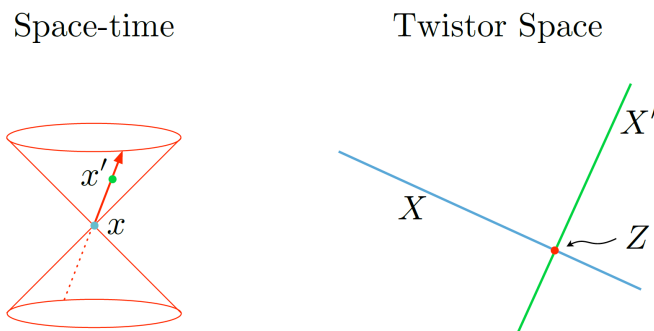


Figure 4: Twistor geometry. Image Credits: [36]

2.4 Real Slices of Twistor Space

So far we have been working with complexified twistor space. In this section, we will see how to get real slices from the complexified twistor space and see if we get a spacetime with a real signature from the twistor incidence relation. Here we provide a general step-by-step procedure to obtain real slices from twistor space:

- Take a real group $\mathcal{G} \subset \mathrm{SO}(6, \mathbb{C})$.
- Restrict twistors to be spinors of $\mathcal{G}$.
- This imposes conditions on ω and π .
- Use the incidence relation to obtain restrictions on spacetime.

If $\mathcal{G}$ is a unitary group then there will be a natural group invariant inner product between ω and π . Note that the existence of spacetime with a real signature is not guaranteed.

Example: $\mathrm{SO}(2, 4)$

$\mathrm{SO}(2, 4)$ is a real subgroup of $\mathrm{SO}(6, \mathbb{C})$. The double cover of $\mathrm{SO}(2, 4)$ is $\mathrm{SU}(2, 2)$. This is a unitary group and has an invariant product on twistors given by:

$$Z^\dagger \gamma_0 Z. \tag{35}$$

Let's investigate this inner product further which is usually written in the following way:

$$Z^\alpha \bar{Z}_\alpha := Z^\dagger \gamma_0 Z \quad (36)$$

$$= \omega^A \bar{\pi}_A + \pi_{\dot{A}} \bar{\omega}^{\dot{A}} \quad (37)$$

$$= ix^{A\dot{A}} \pi_{\dot{A}} \bar{\pi}_A - i \pi_{\dot{A}} \bar{\pi}_A \bar{x}^{\dot{A}A} \quad (38)$$

$$= i(x^{A\dot{A}} - x^{A\dot{A}\dagger}) \pi_{\dot{A}} \bar{\pi}_A. \quad (39)$$

Note that for null twistors the above inner product is zero which implies that the spacetime point is hermitian. Any hermitian matrix can be written in the following way:

$$x = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}, \quad x_i \in \mathbb{R} \quad \forall i. \quad (40)$$

$$\det(x) = x_0^2 - x_1^2 - x_2^2 - x_3^2 \quad (41)$$

Therefore, this corresponds to a spacetime with a Lorentzian signature. The group of isometries is $\text{SO}(1, 3)$ and the conformal group is $\text{SO}(2, 4)$. Hence our calculation is consistent.

Therefore, Lorentzian null twistors correspond to null rays in Minkowski spacetime. In the next two sections when we say twistors we mean Lorentzian twistors. In Appendix B, one can find more examples to obtain real spacetimes with different signatures from twistor space. For a more pedagogical review of twistors, one is referred to [24, 25, 31, 36].

3 Quantized Supertwistors, Superconformal Algebra, and the Penrose Transform

In this section, we will see the algebra of quantized (Lorentzian) twistor space and how the conformal algebra follows from it. We will also introduce the Penrose transform which is an elegant way of writing quantum fields as integrals on twistor space.

3.1 Quantized Twistors and Conformal Algebra

A dual twistor $W_\alpha = (\lambda_A, \mu^{\dot{A}})$ is defined as:

$$W_\alpha : \mathbb{PT} \longrightarrow \mathbb{C} \quad (42a)$$

$$Z^\alpha W_\alpha = \omega^A \lambda_A + \pi_{\dot{A}} \mu^{\dot{A}} \in \mathbb{C}. \quad (42b)$$

We will call the dual twistor space $\mathbb{PT}^*$. Note that eqn. (36) is basically the definition of a dual twistor $\bar{Z}_\alpha = (\bar{\pi}_A, \bar{\omega}^{\dot{A}})$ which gives a $SU(2, 2)$ invariant map from Z^α to $\mathbb{R}$.

We can canonically quantize twistors by defining the following commutation relations:

$$[Z^\alpha, \bar{Z}_\beta] = \delta_\beta^\alpha. \quad (43)$$

The commutation relations $[Z^\alpha, Z_\beta]$ and $[\bar{Z}^\alpha, \bar{Z}_\beta]$ are taken to be trivial. The commutation relation above implies the following non-trivial commutation relations in terms of ω and π :

$$[\omega^A, \bar{\pi}_B] = \delta_B^A \quad (44a)$$

$$[\pi_{\dot{A}}, \bar{\omega}^{\dot{B}}] = \delta_{\dot{A}}^{\dot{B}}. \quad (44b)$$

These commutation relations are physically well-motivated, in fact, it can be shown that these commutation relations imply the conformal algebra:

$$[p_a, p_b] = 0 \quad (45a)$$

$$[M_{ab}, p_c] = i(p_a \eta_{bc} - p_b \eta_{ac}) \quad (45b)$$

$$[M_{ab}, M_{cd}] = i(\eta_{ad} M_{bc} + \eta_{bc} M_{ad} - \eta_{ac} M_{bd} - \eta_{bd} M_{ac}) \quad (45c)$$

$$[K_a, k_b] = 0 \quad (45d)$$

$$[M_{ab}, k_c] = i(k_a \eta_{bc} - k_b \eta_{ac}) \quad (45e)$$

$$[M_{ab}, D] = 0 \quad (45f)$$

$$[D, p_a] = i p_a \quad (45g)$$

$$[D, k_a] = -i k_a \quad (45h)$$

$$[p_a, k_b] = 2i(\eta_{ab} D - M_{ab}) \quad (45i)$$

In order to prove these relations, we first need to describe the notion of quasi-local momentum and angular momentum in twistor theory.

Momentum (null) in twistor theory is defined as the following vector:

$$p_a = \pi_{\dot{A}} \bar{\pi}_A. \quad (46)$$

From the twistor incidence relation, we can see how spacetime translation affects twistor coordinates:

$$\omega^A = i(x^{A\dot{A}} - \delta x^{A\dot{A}})\pi_{\dot{A}} \quad (47a)$$

$$\Rightarrow \omega^A \longrightarrow \omega^A - i\delta x^{A\dot{A}}\pi_{\dot{A}} \quad (47b)$$

$$\Rightarrow \pi_{\dot{A}} \longrightarrow \pi_{\dot{A}}. \quad (47c)$$

Therefore the infinitesimal generator for translation is $-\pi_{\dot{A}}\frac{\partial}{\partial\omega^A}$. If we use the commutation relations given in eqn. (44), then we can replace $-\frac{\partial}{\partial\omega^A}$ with $\bar{\pi}_A$. Therefore the expression for momentum is $\pi_{\dot{A}}\bar{\pi}_A$.

The expression for angular momentum is:

$$M_{ab} = x_ap_b - x_bp_a. \quad (48)$$

If we use the incidence relation and the expression for the momentum, we can derive the expression for the angular momentum as:

$$M_{ab} := M_{A\dot{A}B\dot{B}} = i\epsilon_{\dot{A}\dot{B}}\pi_{(A}\bar{\omega}_{B)} - i\epsilon_{AB}\omega_{(\dot{A}}\bar{\pi}_{\dot{B})}. \quad (49)$$

Following a similar line of reasoning, special conformal transformations and dilation are given by the following expressions:

$$k_a := -\omega_{\dot{A}}\frac{\partial}{\partial\pi^A} = \omega_{\dot{A}}\bar{\omega}_A \quad (50)$$

$$D := \frac{i}{2}(\pi_a\frac{\partial}{\partial\pi^A} - \omega_{\dot{a}}\frac{\partial}{\partial\omega^{\dot{A}}}) = -\frac{i}{2}(\pi_a\bar{\omega}^a - \omega_{\dot{a}}\bar{\pi}^{\dot{a}}). \quad (51)$$

Using these expressions for conformal generators, and the twistor commutation relation eqn. (43), one can verify the conformal algebra eqn. (45). We provide the tedious calculations related to this claim in Appendix C.

The Pauli-Lubanski vector is given by the following expression:

$$S^a = \frac{1}{2}\epsilon^{abcd}M_{bc}p_d. \quad (52)$$

The Pauli-Lubanski vector along with mass squared is the Casimir operator of the Poincaré algebra. For massless particles, it relates to the spin of the particle in the following way:

$$S^a = sp^a. \quad (53)$$

If we plug back the expression for momentum and angular momentum to write the Pauli-Lubanski vector it can be checked that for commuting twistors the spin of the particle turns out to be:

$$s = \frac{1}{2} Z^\alpha \bar{Z}_\alpha. \quad (54)$$

In the case of twistors with non-trivial commutation relations, the spin can be calculated to be:

$$s = \frac{1}{4} (Z^\alpha \bar{Z}_\alpha + \bar{Z}_\alpha Z^\alpha). \quad (55)$$

Making the following replacement:

$$\bar{Z}_\alpha \longrightarrow -\frac{\partial}{\partial Z^\alpha}$$

gives us the following relation:

$$s = \frac{-1}{2} \left(Z^\alpha \frac{\partial}{\partial Z^\alpha} + 2 \right) \quad (56a)$$

$$\Rightarrow -2s - 2 = Z^\alpha \frac{\partial}{\partial Z^\alpha}. \quad (56b)$$

Thus the homogeneity operator is given by $-2s - 2$. In subsection 3.3, we will use this homogeneity operator to define a $-2s - 2$ homogenous, holomorphic function for a particle with spin s and see how the fields with spin s can be written as integrals of this function on twistor space.

3.2 Quantized Supertwistors and Superconformal Algebra

In 1978, Alan Ferber extended Penrose's twistors to $\mathcal{N} = 4$ supertwistors [37]. We have seen that twistor quantum theory gives us the conformal algebra. In this subsection, we will see how to obtain superconformal algebra from quantized supertwistors. Define the supertwistor as:

$$\hat{Z}^\Omega = (Z^\alpha, \Psi^i) \in \mathbb{CP}^{3|4}. \quad (57)$$

Here, $Z^\alpha = (\omega^A, \pi_A)$ are the regular bosonic coordinates and Ψ^i , with $i = 0, 1, 2, 3$ are four Grassmann valued fermionic coordinates. We use capital Greek letters to denote supertwistor indices. $\mathbb{CP}^{3|4}$ simply means that it is

a complex projective space with 4 bosonic and four fermionic coordinates. Z^α being a spinor of the conformal group transforms as a fundamental representation $\mathbf{4}$ of $SU(2, 2)$ whereas Ψ^i transforms as $\bar{\mathbf{4}}$ of $SU(4)_R$. $SU(4)_R$ is the internal R symmetry that acts only in the fermionic sector. We have already seen the conformal group generators in terms of twistor variables. In $\mathcal{N} = 4$, there are 4 additional supercharges that generate the super-Poincaré algebra. These supercharges satisfy the Killing-spinor equation and in terms of twistor variables they are given as:

$$\bar{Q}^{\dot{A}i} = -i\bar{\pi}^{\dot{A}}\Psi^i \quad (58a)$$

$$Q_i^A = i\pi^A\bar{\Psi}_i. \quad (58b)$$

Just as Killing spinors square to give Killing vectors, there are conformal Killing spinors that square to give conformal Killing vectors. They are given as:

$$\bar{S}_{\dot{A}i} = i\omega_{\dot{A}}\bar{\Psi}_i \quad (59a)$$

$$S_i^A = -i\bar{\omega}_A\Psi_i. \quad (59b)$$

The supercharges given by eqn.(58) and eqn.(59) along with eqn.(46, 49, 50), and eqn.(51) generate the $\mathcal{N} = 4$ superconformal algebra that includes the conformal algebra given by eqn.(45) along with:

$$\{Q_A^i, \bar{Q}_B^j\} = 2\delta^{ij}(\sigma^\mu)_{A\dot{B}}p_\mu \quad (60a)$$

$$\{Q_A^i, Q_B^j\} = \{\bar{Q}_A^i, \bar{Q}_B^j\} = 0 \quad (60b)$$

$$\{S_A^i, \bar{S}_B^j\} = 2\delta^{ij}(\sigma^\mu)_{A\dot{B}}k_\mu \quad (60c)$$

$$\{S, S\} = \{\bar{S}, \bar{S}\} = 0 \quad (60d)$$

$$\{Q, \bar{S}\} = \{\bar{Q}, S\} = 0 \quad (60e)$$

$$\{Q_A^i, S_B^j\} = \epsilon_{AB}(\delta^{ij}D + T^{ij}) + \frac{1}{2}\delta^{ij}M_{\mu\nu}(\sigma^{\mu\nu})_{AB}. \quad (60f)$$

Here $(\sigma^{\mu\nu})_{AB} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]^{AB}$ are the Lorentz generators and T^{ij} is some adjoint matrix in $SU(4)$. The fermionic, bosonic interaction part of the algebra is

given by:

$$[Q, p] = [\bar{Q}, p] = 0 \quad (61a)$$

$$[M^{\mu\nu}, Q_A^i] = \frac{1}{2}(\sigma^{\mu\nu})_A^B Q_B^i \quad (61b)$$

$$[S, k] = [\bar{S}, k] = 0 \quad (61c)$$

$$[D, Q] = \frac{i}{2}Q, \quad [D, \bar{Q}] = \frac{i}{2}\bar{Q} \quad (61d)$$

$$[D, S] = \frac{-i}{2}S, \quad [D, \bar{S}] = \frac{-i}{2}\bar{S}. \quad (61e)$$

As can be seen, the superconformal algebra is huge but it can be obtained elegantly from quantized supertwistors. Let us first define the dual of a supertwistor:

$$\bar{\hat{Z}}_\Omega := (\bar{Z}_\alpha, \bar{\Psi}_i) \quad (62a)$$

$$= (\bar{\pi}_A, \bar{\omega}^{\dot{A}}, \bar{\Psi}_i). \quad (62b)$$

The dual supertwistor is simply the charge conjugate of the supertwistor with ω and π swapped. Therefore the inner product between a supertwistor and a dual supertwistor will have a $SU(2, 2)$ invariant bosonic part and a $SU(4)$ invariant fermionic part.

$$\hat{Z}^\Omega \bar{\hat{Z}}_\Omega = \omega^A \pi_A + \pi_{\dot{A}} \omega^{\dot{A}} + \Psi^i \bar{\Psi}_i \quad (63)$$

We perform deformation quantization of supertwistor space by introducing the following (anti-)commutation relations:

$$[Z^\alpha, \bar{Z}_\beta] = \delta_\beta^\alpha, \quad [Z^\alpha, Z_\beta] = [\bar{Z}^\alpha, \bar{Z}_\beta] = 0 \quad (64a)$$

$$\{\Psi^i, \bar{\Psi}_j\} = \delta_j^i, \quad \{\Psi^i, \Psi_j\} = \{\bar{\Psi}^i, \bar{\Psi}_j\} = 0 \quad (64b)$$

$$[Z^\alpha, \Psi^i] = [Z^\alpha, \bar{\Psi}^i] = [\bar{Z}^\alpha, \Psi^i] = 0. \quad (64c)$$

The above relations can be elegantly written using the super-bracket in the following way:

$$[\hat{Z}^\Omega, \bar{\hat{Z}}_\Theta]_\mp = \delta_\Theta^\Omega, \quad [\hat{Z}^\Omega, \hat{Z}^\Theta]_\mp = 0, \quad [\bar{\hat{Z}}_\Omega, \bar{\hat{Z}}_\Theta]_\mp = 0. \quad (65)$$

It can be shown that if we use the expression for superconformal generators in terms of supertwistor variables and impose the above (anti-)commutation rules then the superconformal algebra follows naturally (Check Appendix C and D).

3.3 Penrose Transform

The Penrose transform provides a way of expressing wave functions of massless fields as contour integrals of homogeneous holomorphic functions on twistor space [27].

Let's take the example of source-free Maxwell's equation. The anti-symmetric field strength tensor can be decomposed in the following way:

$$F_{ab} = F_{A\dot{A}B\dot{B}} = \epsilon_{AB}\tilde{F}_{\dot{A}\dot{B}} + \epsilon_{\dot{A}\dot{B}}F_{AB}. \quad (66)$$

Here, $\tilde{F}_{\dot{A}\dot{B}}$ and F_{AB} are symmetric in their spinor indices and are called the self-dual and anti-self dual decomposition of the field strength tensor respectively.

The Maxwell equation reduces to the following form:

$$\partial^a F_{ab} = 0 \quad (67a)$$

$$\Rightarrow \partial_B^{\dot{A}}\tilde{F}_{\dot{A}\dot{B}} + \partial_{\dot{B}}^A F_{AB} = 0. \quad (67b)$$

Similarly, the Bianchi identity can be written in the following way:

$$\frac{1}{2}\epsilon^{abcd}\partial_b F_{cd} = 0 \quad (68a)$$

$$\Rightarrow \partial_B^{\dot{A}}\tilde{F}_{\dot{A}\dot{B}} - \partial_{\dot{B}}^A F_{AB} = 0. \quad (68b)$$

Combining eqn. (67b) and eqn. (68b), we can write the following equations:

$$\partial_B^{\dot{A}}\tilde{F}_{\dot{A}\dot{B}} = 0 \quad (69a)$$

$$\partial_{\dot{B}}^A F_{AB} = 0. \quad (69b)$$

Eqn. (69a) and (69b) are called the self-dual and anti-self dual Maxwell's equation respectively.

Generalizing to the case of higher spin we get the following equations of motion:

$$\partial^{A\dot{A}}\psi_{ABC\dots F} = 0 \quad \text{and} \quad \partial^{A\dot{A}}\tilde{\psi}_{\dot{A}\dot{B}\dots\dot{F}} = 0. \quad (70)$$

$A, B, \dots F$ being $2s$ in number where s is the particle's spin. For spinless particles, we have the massless Klein-Gordon equation given by:

$$\partial_{A\dot{A}}\partial^{A\dot{A}}\phi = 0. \quad (71)$$

The claim is that the solution of eqn. (70) and (71) can be written using a holomorphic ‘twistor function’ $f(Z^\alpha)$ of homogeneity $-2s - 2$ and -2 (for spin 0) respectively. The holomorphicity and homogeneity conditions are:

$$\frac{\partial f}{\partial \bar{Z}_\alpha} = 0 \quad Z^\alpha \frac{\partial f}{\partial Z^\alpha} = (-2s - 2)f \quad (72)$$

The solution of equation (70) and (71) are given by:

$$\tilde{\psi}_{\dot{A}\dot{B}\dot{C}\dots\dot{F}} = \frac{1}{2\pi i} \oint_{\omega=ix\pi} \pi_{\dot{A}}\pi_{\dot{B}}\dots\pi_{\dot{F}} f(Z^\alpha) \pi_{\dot{Y}} d\pi^{\dot{Y}}. \quad (73)$$

The above holds true for $s \geq 0$ and for $s \leq 0$ we have the following:

$$\psi_{AB\dots F} = \frac{1}{2\pi i} \oint_{\omega=ix\pi} \frac{\partial}{\partial \omega^A} \frac{\partial}{\partial \omega^B} \dots \frac{\partial}{\partial \omega^F} f(Z^\alpha) \pi_{\dot{Y}} d\pi^{\dot{Y}}. \quad (74)$$

It can be easily verified that the solutions given by eqn.(73) and eqn.(74) satisfy eqn.(69a) and eqn.(69b) for a particle with spin s . For a spinless particle the solutions given by eqn.(73) and eqn.(74) satisfy eqn.(71). This can be checked by taking the derivative in eqn.(69a, 69b, 71) inside the integral and showing that it vanishes. Using the incidence relation, the spacetime derivative is given by:

$$\partial_{A\dot{A}} = i\pi_{\dot{A}} \frac{\partial}{\partial \omega^A}. \quad (75)$$

In this section, we saw how twistor space is a space where massless quantum fields live. Massless quantum fields are contour integrals of homogenous holomorphic functions in twistor space. Is there any way to understand gauge symmetry groups associated with these quantum fields on twistor space?

4 Quantized Supertwistors and (Split-)Octonions

In this section, we will see how to obtain the algebra of (split-)octonions from quantized twistors following Penrose’s work in [1]. We will further explore the quantized space of $\mathcal{N} = 4$ supertwistors and claim to get both the octonions and split-octonions. The algebra of (split-)octonions can then be used to construct symmetry groups that act on fields on the twistor space. We discuss one such construction in section 5.

4.1 Quantized Twistors and Split-Octonions

A bitwistor is an 8-component object given by a twistor and dual twistor (Z^α, W_α) . Quantized bitwistors satisfy the following commutation relations:

$$Z^\alpha W_\beta - W_\beta Z^\alpha = \delta_\beta^\alpha, \quad (76)$$

together with,

$$Z^\alpha Z^\beta = Z^\beta Z^\alpha, \quad W_\alpha W_\beta = W_\beta W_\alpha. \quad (77)$$

Z^α and W_α together make the basis of a quantized bitwistor algebra given by $\mathcal{O}$. An element of this algebra is given by:

$$\mathcal{A} = \bar{A}_\alpha Z^\alpha + F^\alpha W_\alpha \in \mathcal{O}. \quad (78)$$

Here $\bar{A}_\alpha$ and F^α are $\mathbb{C}$ -valued commuting coefficients. Note that we are inheriting the $\mathfrak{su}(2, 2)$ invariant twistor inner product $(\bar{A}_\alpha Z^\alpha)$ to define this algebra. For the algebra to be over the field of $\mathbb{R}$, we have the following reality constraints:

$$F^\alpha = A^\alpha, \quad W_\alpha = \bar{Z}_\alpha. \quad (79)$$

For this to be an algebra it should have a well-defined bilinear product. In order to define that we first introduce the following triple product:

$$[\mathcal{A}, \mathcal{B}, \mathcal{C}] = i \sum_{\sigma} (-1)^{|\sigma|} \sigma(\mathcal{A}\mathcal{B}\mathcal{C}), \quad (80)$$

for any three elements $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C} \in \mathcal{O}$ given by:

$$\mathcal{A} = \bar{A}_\alpha Z^\alpha + A^\alpha \bar{Z}_\alpha, \quad \mathcal{B} = \bar{B}_\alpha Z^\alpha + B^\alpha \bar{Z}_\alpha, \quad \mathcal{C} = \bar{C}_\alpha Z^\alpha + C^\alpha \bar{Z}_\alpha. \quad (81)$$

Here, σ gives all permutations of $\mathcal{A}\mathcal{B}\mathcal{C}$ and $(-1)^{|\sigma|}$ gives -1 for all odd permutations. The triple product simplifies to the following form:

$$\sum_{\circlearrowleft} i\mathcal{A}(\bar{B}_\beta C^\beta - \bar{C}_\beta B^\beta). \quad (82)$$

The other two terms correspond to $\mathcal{B}$ and $\mathcal{C}$ in a similar cyclic manner.

Scalar multiplication is given in the following trivial way, for $\nu \in \mathbb{R}$:

$$\nu\mathcal{A} := (\nu\bar{A}_\alpha)Z^\alpha + (\nu A^\alpha)\bar{Z}_\alpha \in \mathcal{O}. \quad (83)$$

Inner product is a map from $(\mathcal{A}, \mathcal{B}) \in \mathcal{O} \times \mathcal{O}$ to $\mathcal{A} \cdot \mathcal{B} \in \mathbb{R}$ defined in the following way:

$$\mathcal{A} \cdot \mathcal{B} := i\mathcal{A}(\mathbf{I}\mathcal{B}) - i(\mathbf{I}\mathcal{B})\mathcal{A} \quad (84a)$$

$$= (\bar{A}_\alpha B^\alpha + A^\alpha \bar{B}_\alpha) \in \mathbb{R} \quad (84b)$$

$$= |A^0 + B^2|^2 + |A^1 + B^3|^2 + |A^2 + B^0|^2 + |A^3 + B^1|^2 - |A^0 - B^2|^2 - |A^1 - B^3|^2 - |A^2 - B^0|^2 - |A^3 - B^1|^2 \quad (84c)$$

Where $\mathbf{I}$ is defined as:

$$\mathbf{I} : \bar{A}_\alpha Z^\alpha + A^\alpha \bar{Z}_\alpha \longrightarrow i(\bar{A}_\alpha Z^\alpha - A^\alpha \bar{Z}_\alpha) \in \mathcal{O}. \quad (85)$$

One can obtain eqn. (84b) from eqn. (84a) by using the commutation relations given by eqn.(43). Note that the inner product in eqn. (84c) has a (4,4) split-signature. This indicates that the algebra $\mathcal{O}$ might be that of the split-octonions.

If we choose a unit bitwistor:

$$\mathcal{E} = \bar{E}_\alpha Z^\alpha + E^\alpha \bar{Z}_\alpha, \quad (86)$$

such that

$$\mathcal{E} \cdot \mathcal{E} = 2, \quad (87)$$

then we can do the following decomposition of any element in the bitwistor algebra:

$$\mathcal{A} = \mathcal{A}^S + \mathcal{A}^V. \quad (88)$$

Where,

$$\mathcal{A}^S = \frac{1}{2}(\mathcal{A} \cdot \mathcal{E})\mathcal{E} \quad \text{and} \quad \mathcal{A}^V \cdot \mathcal{E} = 0. \quad (89)$$

The unit element $\mathcal{E}$ is not uniquely defined. The decomposition along $\mathcal{E}$ will be the real part of the algebra whereas the decomposition perpendicular to $\mathcal{E}$ will be the imaginary part. This geometric-algebra-inspired decomposition motivates the following bilinear product between two elements of the bitwistor algebra:

$$\begin{aligned} \mathcal{A} \vee \mathcal{B} &= (\mathcal{A}^S + \mathcal{A}^V) \vee (\mathcal{B}^S + \mathcal{B}^V) \\ &= (\mathcal{A}^S \cdot \mathcal{B}^S - \mathcal{A}^V \cdot \mathcal{B}^V)\mathcal{E} + ((\mathcal{A} \cdot \mathcal{E})\mathcal{B}^V + (\mathcal{B} \cdot \mathcal{E})\mathcal{A}^V + \mathcal{A}^V \times \mathcal{B}^V) \in \mathcal{O}. \end{aligned} \quad (90)$$

Where the cross-product $\mathcal{A} \times \mathcal{B}$ is given by:

$$\mathcal{A} \times \mathcal{B} = [\mathbf{I}\mathcal{A}, \mathbf{I}\mathcal{B}, \mathbf{I}\mathcal{E}]. \quad (91)$$

The bilinear product given in eqn. (90) is not associative but is alternative, i.e. $\mathcal{A} \vee (\mathcal{A} \vee \mathcal{B}) = (\mathcal{A} \vee \mathcal{A}) \vee \mathcal{B}$. It can be further checked that this is a composition algebra in the following way. If we define complex conjugation in the following way:

$$\mathcal{A}^* = \mathcal{A}^S - \mathcal{A}^V, \quad (92)$$

then it is easy to check that $\mathcal{A} + \mathcal{A}^*$ and $|\mathcal{A}|^2 = \mathcal{A} \cdot \mathcal{A}^* = \mathcal{A}^* \cdot \mathcal{A}$ are along $\mathcal{E}$.

Therefore, we now have an alternative, non-associative composition algebra that has a $(4, 4)$ split signature. This is the algebra of the split-octonions.

4.2 Quantized Supertwistors and (Split-)Octonions

We have seen that twistor quantum theory gives us the algebra of split-octonions and the conformal algebra. We have also seen that quantized supertwistors give us the superconformal algebra. What should we expect if we repeat the analysis in the previous subsection (4.1) for supertwistors?

Let us see first, what happens on quantizing the fermionic $SU(4)$ spinor part of the supertwistor. Let $\Psi^i = (\Psi^0, \Psi^1, \Psi^2, \Psi^3)$ be a spinor of $SU(4)$. Then the dual spinor is given by $\bar{\Psi}_i = (\bar{\Psi}_0, \bar{\Psi}_1, \bar{\Psi}_2, \bar{\Psi}_3)$ such that the inner product:

$$\Psi^i \bar{\Psi}_i = |\Psi^0|^2 + |\Psi^1|^2 + |\Psi^2|^2 + |\Psi^3|^2, \quad (93)$$

is invariant under $SU(4)$. A real quantized bispinor algebra is defined in a similar way:

$$\mathcal{A} = \bar{A}_i \Psi^i + A^i \bar{\Psi}_i \in \mathcal{O}. \quad (94)$$

A^i is a $\mathbb{C}$ -valued coefficient such that $A^i \bar{\Psi}_i$ is $SU(4)$ invariant. The quantum relations between Ψ^i and $\bar{\Psi}^i$ are given by:

$$\{\Psi^i, \bar{\Psi}_j\}_+ = \delta_j^i \quad (95a)$$

$$\{\Psi^i, \Psi^j\}_+ = 0 \quad (95b)$$

$$\{\bar{\Psi}_i, \bar{\Psi}_j\}_+ = 0. \quad (95c)$$

Define the following inner products:

$$\langle \mathcal{A} \mathcal{B} \rangle := \mathcal{A} \mathcal{B} + \mathcal{B} \mathcal{A} \quad (96a)$$

$$= \bar{A}_i B^i + A^i \bar{B}_i \quad (96b)$$

Eqn.(96b) follows from eqn.(96a) using the anti-commutation rules given by eqn.(95a). This inner product has an $(8, 0)$ non-split signature which is

indicative of octonion algebra. We can obtain the bilinear product of the octonionic algebra by defining the following triple product:

$$\sum_{cyclic} [(IA)B + B(IA)]C \in \mathcal{O}, \quad (97)$$

which reduces to a similar form as in eqn.(82):

$$[\mathcal{A}, \mathcal{B}, \mathcal{C}] = \sum_{cyclic} i(A_i B^i - A^i B_i)(\mathcal{C}) \in \mathcal{O}. \quad (98)$$

The rest of the analysis follows from the commuting case where we define an identity of the algebra and the octonionic bilinear product is the triple product of two octonions along with the identity.

Therefore, we see that quantizing the fermionic part of the supertwistors gives us the octonions just as quantizing the bosonic part gives us the split-octonions. What happens if we repeat the analysis for the quantized supertwistor without breaking it separately into fermionic and bosonic subparts? In that case, the algebra obtained is the superalgebra $\tilde{\mathcal{O}} \oplus \mathcal{O}$; the calculations for this claim can be found in Appendix E.

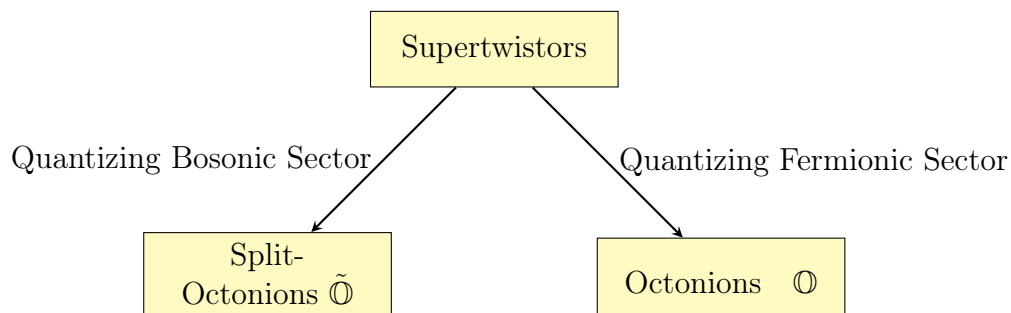


Figure 5: Choosing an identity in the quantized supertwistor algebra breaks the symmetry from $\mathfrak{su}(2, 2|4)$ to the Lie algebra of the automorphism group of octonions $\mathfrak{g}_2$ and split-octonions $\mathfrak{g}_2^*$ respectively.

So far, we have a mathematical result. Quantized $\mathcal{N} = 4$ supertwistors give us the superconformal, split octonionic, and octonionic algebras. What is it good for? Split octonions and Octonions have just the right number of degrees of freedom to explain the Standard Model gauge group along with Lorentz symmetry. In the next section, we review a GUT based on (split-) octonionic model [2].

5 Grand Unified Theories with Split-Octonions and Octonions

This section will be a review of [2], where the author has shown how to use (split-)octonions to construct the kinematical symmetry (that includes Lorentz symmetry) of a prospective GUT. We start with octonions to see the construction of $\text{Spin}(7)$, we then do something similar with split-octonions to get $\text{Spin}(4, 3)$, and finally, we see the construction of $\text{Spin}(11, 3)$ GUT with $\mathbb{O} \otimes \mathbb{O}$.

5.1 Octonions and $\text{Spin}(7)$

The algebra of the octonions is non-associative. However, if we define the left multiplication by octonions as a series of nested maps then this is associative since all maps are associative. Let us denote L_{e_i} as the left multiplication action by the imaginary octonion e_i and consider the endomorphism given by the operators:

$$E^i := L_{e_i}. \quad (99)$$

Since any two imaginary octonions anti-commute and square to -1 (refer to Fig.1), we have:

$$E^i E^j + E^j E^i = -2\delta^{ij}. \quad (100)$$

There are 7 imaginary octonions, however, the 7th operator E^7 can be written as $E^1 E^2 E^3 E^4 E^5 E^6$ and therefore the Clifford algebra obtained is $Cl(7)^- \cong Cl(6)$.

If we define the imaginary octonions as the odd-graded generators of $Cl(7)$ then we can write the generators of the Lie algebra $\mathfrak{spin}(7)$ as:

$$\omega_{ij} = \frac{-1}{4} [E_i, E_j]. \quad (101)$$

Therefore, imaginary octonions are the spinor representation of $\text{SO}(7)$:

$$\text{Im}(\mathbb{O}) \cong S_7. \quad (102)$$

There is an interesting choice of complex structure in $\mathfrak{spin}(7)$ given by $J = L_{e_4}$ (left multiplication by e_4). The commutant of this complex structure in

$\mathfrak{spin}(7)$ is $\mathfrak{su}(3) \times \mathfrak{u}(1)$. This can be seen in the following way. Consider a general octonion:

$$\omega = a_0 + a_1 e_1 + a_2 e_2 + a_3 e_3 + a_4 e_4 + a_5 e_5 + a_6 e_6 + a_7 e_7. \quad (103)$$

This can be written as:

$$\omega = (a_0 + a_4 e_4) + (a_1 - a_5 e_4) e_1 + (a_2 - a_6 e_4) e_2 + (a_3 - a_7 e_4) e_3. \quad (104)$$

Therefore the choice of this complex structure breaks $\omega \in \mathbb{O}$ to $\mathbb{C} \oplus \mathbb{C}^3$ with

$$a_0 + a_4 e_4 \in \mathbb{C}, \quad \begin{pmatrix} a_1 - a_5 e_4 \\ a_2 - a_6 e_4 \\ a_3 - a_7 e_4 \end{pmatrix} \in \mathbb{C}^3. \quad (105)$$

If we had chosen the complex structure $J' = R_{e_4}$ then the commutant would have been $\mathfrak{spin}(6) \cong \mathfrak{su}(4)$. This would have rotated all the directions except for the one corresponding to E_4 . However, choosing L_{e_4} results in an additional negative sign in the imaginary coefficients in $\mathbb{C}^3$ without giving a minus sign to the imaginary coefficient in $\mathbb{C}$ eqn.(105) (refer to Fig. 1), thus the symmetry $\mathfrak{spin}(7)$ is further broken to $\mathfrak{su}(3) \times \mathfrak{u}(1)$. Many authors have associated this commutant of J in $\text{Spin}(7)$ with the broken $\text{SU}(3)_c \times \text{U}(1)_{em}$ group of the Standard Model with the quarks represented in $\mathbb{C}^3$ and the leptons represented in $\mathbb{C}$ [10, 12].

5.2 Split-Octonions and $\text{Spin}(4,3)$

We can play the same game with split-octonions by introducing the left-multiplication maps $L_{\tilde{e}_i}$ and their corresponding operators $\tilde{E}^i$. The Clifford algebra generated in this case will be:

$$\tilde{E}^i \tilde{E}^j + \tilde{E}^j \tilde{E}^i = -2\eta^{ij}. \quad (106)$$

There are two possible Clifford algebras that we can obtain. If we take $\tilde{E}_7$ that squares to +1 to be the product of other operators then the Clifford algebra obtained will be $Cl(3,3)$. However, if take the dependent matrix to be $\tilde{E}_1$ that squares to -1, the Clifford algebra we obtain will be $Cl(4,2)$.

Defining the imaginary split octonions as the odd-graded generators of $Cl(4,3)$ we can generate the Lie algebra $\mathfrak{spin}(4,3)$:

$$\omega_{ij} = \frac{-1}{4} [\tilde{E}_i, \tilde{E}_j]. \quad (107)$$

Therefore, we establish the fact:

$$\mathrm{Im}(\tilde{\mathbb{O}}) \cong S_{4,3}. \quad (108)$$

A natural choice of complex structure is $\tilde{J}' = R_{\tilde{e}_4}$. The commutant of this in $\mathfrak{spin}(4, 3)$ is $\mathfrak{spin}(4, 2) \cong \mathfrak{su}(2, 2)$. This rotates all the directions while keeping the direction along $\tilde{E}_4$ invariant. This is interesting because the commutant is the double cover of the conformal group of Minkowski space.

Another choice of complex structure is $\tilde{J} = \tilde{E}_5 \tilde{E}_6 \tilde{E}_7$. Each of $\tilde{E}_i, i = 5, 6, 7$ square to $+1$ but it can be checked that $\tilde{J}^2 = -1$. The commutant of this complex structure in $\mathfrak{spin}(4, 3)$ is $\mathfrak{spin}(3) \times \mathfrak{spin}(1, 3)$. This can be understood as the action of $\mathfrak{spin}(3)$ on the vectors $\tilde{E}_5, \tilde{E}_6, \tilde{E}_7$ in their plane whereas $\mathfrak{spin}(1, 3)$ rotates the other vectors separately.

In [2], this choice of complex structure is taken to be of physical relevance since the group $\mathrm{Spin}(1, 3)$ is identified with the Lorentz group and $\mathfrak{spin}(3) \cong \mathfrak{su}(2)$ is identified with the Weak group in the Standard Model.

5.3 $\mathrm{Spin}(11, 3)$ Unification

The Standard Model Weyl fermions have 16 complex off-shell degrees of freedom. If we consider the spinor indices, then they have 32 complex or 64 real degrees of freedom. Therefore, if we have to propose a group that has both Standard Model and spacetime symmetries, then the spinors of that group should have 64 real degrees of freedom. It is well known that spinors of $Cl(p, q)$ with $p + q = 2n$ have 2^n degrees of freedom, moreover, the spinors are Majorana (real) if $p - q = 8n$. Thus, we see that the group $\mathrm{Spin}(11, 3)$ satisfies all the required conditions of having Majorana-Weyl spinors that have 64 real degrees of freedom. Using $\mathrm{Spin}(11, 3)$ for explaining both gauge and gravity was first proposed by Percacci in [38]. Recent work includes the work of Kirill Krasnov [2]. In this section, we briefly review the work by Krasnov.

Consider matrices taking values in $\mathrm{End}((\mathbb{O} \otimes \tilde{\mathbb{O}})^2)$:

$$\Gamma_x = \begin{pmatrix} 0 & L_x \otimes \mathbb{I} \\ L_{\bar{x}} \otimes \mathbb{I} & 0 \end{pmatrix}, x \in \mathrm{Im}(\mathbb{O}), \quad \Gamma_y = \begin{pmatrix} 0 & -\mathbb{I} \otimes L_y \\ \mathbb{I} \otimes L_{\bar{y}} & 0 \end{pmatrix}, y \in \mathrm{Im}(\tilde{\mathbb{O}}). \quad (109)$$

Here L_x and L_y are operators of left multiplication by $x \in \mathrm{Im}(\mathbb{O})$ and $y \in \mathrm{Im}(\tilde{\mathbb{O}})$ respectively. $\bar{x}$ and $\bar{y}$ are conjugation. These matrices satisfy the

following algebra:

$$\Gamma_x \Gamma_y + \Gamma_y \Gamma_x = 0 \quad (110a)$$

$$\Gamma_x^2 = |x|^2 \mathbb{I}, \quad \Gamma_y^2 = -|y|^2 \mathbb{I}. \quad (110b)$$

Thus, these matrices generate the Clifford algebra $Cl(11, 3)$ with $Cl(7, 0)$ coming from the octonions and $Cl(4, 3)$ coming from the split octonions (notice the minus sign in front of $|y|^2$ in 110a). Using these matrices, we can generate the Lie algebra of $\mathfrak{spin}(11, 3)$.

$\text{Spin}(11, 3)$ contains $\text{Spin}(10) \times \text{Spin}(1, 3)$ as a maximal subgroup, where $\text{Spin}(10)$ gives the internal gauge symmetry and $\text{Spin}(1, 3)$ gives the Lorentz symmetry. It is a well-established fact that the Weyl spinor of $\text{Spin}(10)$ contains all the fermion representations of the left-right symmetric Standard Model. A Majorana-Weyl spinor of $\text{Spin}(11, 3)$ can be written as:

$$S_{(11,3)}^+ = S_{10}^+ \otimes S_{1,3}^+ \oplus S_{10}^- \otimes S_{1,3}^-. \quad (111)$$

This is precisely what we need for a Standard Model Weyl Fermion equipped with Lorentz degree of freedom.

How does $\text{Spin}(11, 3)$ break into the gauge groups and Lorentz group? Recall the complex structures J and $\tilde{J}$ introduced in sections 5.1 and 5.2 respectively. The commutant of J in $\mathfrak{spin}(7)$ is $\mathfrak{su}(3) \oplus \mathfrak{u}(1)$, whereas the commutant of $\tilde{J}$ in $\mathfrak{spin}(4, 3)$ is $\mathfrak{su}(2) \oplus \mathfrak{spin}(1, 3)$. To understand the commutant of $J, \tilde{J}$ acting on $\mathbb{O} \otimes \tilde{\mathbb{O}}$, note that $E^4 \otimes \tilde{E}^i, i = 5, 6, 7$ commutes with both J and $\tilde{J}$. Therefore, the commutant $\mathfrak{spin}(3)$ of $\tilde{J}$ in $\mathfrak{spin}(4, 3)$ gets extended to the commutant $\mathfrak{spin}(4)$ of J and $\tilde{J}$ in $\mathfrak{spin}(11, 3)$ causing rotations among the four axis $E^4, \tilde{E}^i, i = 5, 6, 7$. The overall commutant of J and $\tilde{J}$ in $\mathfrak{spin}(11, 3)$ is the Lie algebra of the group:

$$G = \text{SU}(3)_c \times \text{SU}(2)_L \times \text{SU}(2)_R \times \text{U}(1)_Y \times \text{Spin}(1, 3). \quad (112)$$

$\text{Spin}(4)$ has been written as $\text{SU}(2)_L \times \text{SU}(2)_R$. The lower indices have been added to familiarize the above group with the Left-Right symmetric gauge group tensor Lorentz group.

Note that in section 5.1 we also introduced the complex structure J' whose commutant in $\mathfrak{spin}(7)$ is $\mathfrak{spin}(6) \cong \mathfrak{su}(4)$. If we choose J' and $\tilde{J}$ as complex structures on $\mathfrak{spin}(11, 3)$ then the Lie group corresponding to the commutant of this in $\mathfrak{spin}(11, 3)$ is the Pati-Salam gauge group:

$$G = \text{SU}(4) \times \text{SU}(2)_L \times \text{SU}(2)_R \times \text{U}(1)_Y. \quad (113)$$

Therefore, we see that 2×2 matrices with entries in $\mathbb{O} \otimes \tilde{\mathbb{O}}$ can act on $(\mathbb{O} \otimes \tilde{\mathbb{O}})^2$ under the symmetry $\text{Spin}(11, 3)$ that geometrically breaks into $\mathcal{G}_{\text{Gauge}} \times \text{Spin}(1, 3)$.

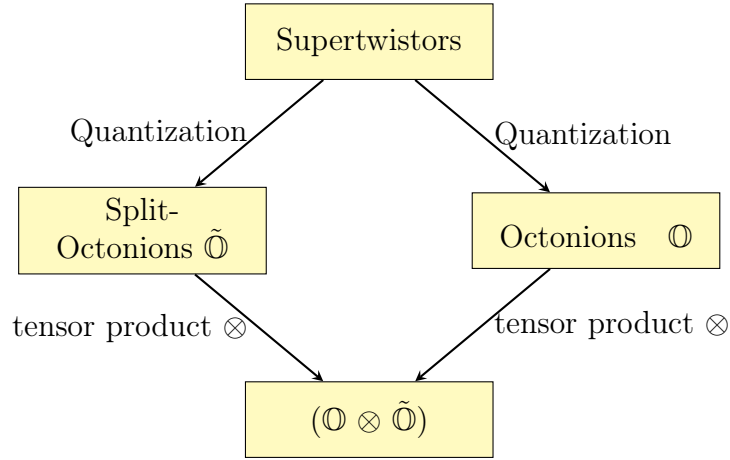


Figure 6: Choosing an identity breaks the quantized supertwistor algebra into the algebra of octonions and split-octonions. Doing a tensor product of these algebras gives $(\mathbb{O} \otimes \tilde{\mathbb{O}})$ that transforms as a Majorana-Weyl spinor of $\text{Spin}(11, 3)$ and contains all the Standard Model fermions with Lorentz symmetry.

6 Conclusion and Future Work

We note that deformation quantization of $\mathcal{N} = 4$ supertwistor space and choosing the identity $\mathcal{E}$ breaks the superconformal algebra $\mathfrak{su}(2, 2|4)$ to the Lie algebra of the automorphism groups of octonions ($\mathfrak{g}_2$) and split octonions ($\mathfrak{g}_2^*$). This symmetry breaking is algebraic and is brought about by the choice of identity $\mathcal{E}$ in the algebra. The construction of $\text{Spin}(11, 3)$ is an example of obtaining minimalistic Standard Model physics from octonions and split octonions. We will further investigate if there is a more natural construction that follows from quantized supertwistors and produces the Standard Model algebra with Lorentz symmetry. This project was limited to the algebraic analysis of various symmetry groups but we haven't discussed any dynamics. Usually, grand unified groups that have the Lorentz group as a subgroup are ruled out by the Coleman-Mandula theorem which states that non-trivial representations of internal gauge groups should be Lorentz scalars [39]. There

are ways to get around the Coleman-Mandula theorem, supersymmetry being the most famous one. In this project, the starting point is $\mathcal{N} = 4$ supertwistor theory and there is only the superconformal group which eventually breaks into the (split-)octonionic algebra.

There are many interesting things related to this project that are currently work in progress. We will very briefly mention some of these ideas here without presenting rigorous arguments.

The non-compact group $E_{8(8)}$ has the following triality decomposition:

$$E_{8(8)} = \text{Tri}(\mathbb{O}) \oplus \text{Tri}(\tilde{\mathbb{O}}) \oplus (\mathbb{O} \otimes \tilde{\mathbb{O}}) \oplus (\mathbb{O} \otimes \tilde{\mathbb{O}}) \oplus (\mathbb{O} \otimes \tilde{\mathbb{O}}). \quad (114)$$

The triality algebra of the octonions, $\text{Tri}(\mathbb{O})$ is $\mathfrak{spin}(8)$ and the triality algebra of the split-octonions, $\text{Tri}(\tilde{\mathbb{O}})$ is $\mathfrak{spin}(4, 4)$. The algebra $\text{Tri}(\mathbb{O}) \oplus \text{Tri}(\tilde{\mathbb{O}}) \oplus \mathbb{O} \otimes \tilde{\mathbb{O}}$ is isomorphic to $\mathfrak{spin}(12, 4)$ and there are three ways to obtain this by choosing any one of the three $\mathbb{O} \otimes \tilde{\mathbb{O}}$ appearing in eqn.(114). The remaining part of the algebra which is $\mathbb{O} \otimes \tilde{\mathbb{O}} \oplus \mathbb{O} \otimes \tilde{\mathbb{O}}$ transforms as a Majorana-Weyl spinor of $\text{Spin}(12, 4)$. $\text{Spin}(12, 4)$ is the conformal group of $\text{Spin}(11, 3)$ and the algebra naturally splits into $\mathfrak{spin}(10) \oplus \mathfrak{spin}(2, 4)$ which is the $\text{SO}(10)$ GUT with the conformal group of Minkowski space. We have seen in section 5.3 that all the Standard Model fermions with Lorentz spin fit inside $\mathbb{O} \otimes \tilde{\mathbb{O}}$. Since there are three ways of choosing $\mathbb{O} \otimes \tilde{\mathbb{O}}$ to obtain $\mathfrak{spin}(12, 4)$, it might be a mathematical explanation behind the existence of three generations of fermions. In [12], this idea has already been presented in terms of the triality decomposition of E_6 . In the case of E_6 however, we get three generations of fermions transforming under $\text{SO}(10)$ GUT without Lorentz symmetry. Using $E_{8(8)}$ is a natural extension of that idea to include Lorentz symmetry. Also, note the fact that $\mathbb{O} \otimes \tilde{\mathbb{O}}$ can be thought of as a tensor product between quantized bosonic and fermionic supertwistor algebra.

Another interesting research direction motivated purely out of mathematical curiosity is to see the algebra of quantized $\mathcal{N} = 1$, and $\mathcal{N} = 2$ supertwistors. We have not shown in this essay explicitly but it seems that quantized supertwistors give a series of superalgebras $\tilde{\mathbb{O}} \oplus \mathbb{C}$, $\tilde{\mathbb{O}} \oplus \mathbb{H}$, and $\tilde{\mathbb{O}} \oplus \mathbb{O}$ corresponding to $\mathcal{N} = 1, 2$, and 4 respectively.

To summarize, this project is far from being complete and there are many aspects that need to be understood properly. Even though it is a mathematical result that quantized supertwistors give us the algebra of (split-) octonions, its physical significance is not very clear. There are models based on (split-)octonions that explain the Standard Model with Lorentz symmetry

but it needs to be seen how they naturally emerge from supertwistors. We need to make testable predictions and explain the unexplained mysteries of the Standard Model such as the existence of three generations of fermions. Various GUT models are ruled out because of extra gauge fields that lead to proton decay but the internal gauge group we are proposing is $SO(10)$, and there are $SO(10)$ models with the decay rate for protons within the experimental bound [40, 41]. The $Spin(11, 3)$ GUT breaks into a left-right symmetric gauge model. These models predict a right-handed neutrino that could be a cold dark matter candidate [42, 43], explain neutrino mass using a see-saw mechanism [44, 45], and explain the conservation of charge-parity in the strong interaction and the vanishing of Higgs coupling prior to Electroweak breaking [46, 47]. These predictions based on minimalistic assumptions make it very tempting to explore the underlying mathematics of these models.

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A Lie Algebras, Jordan Algebras, and the Magic Square

Lie algebra is the algebra of generators of a continuous group (a.k.a. Lie group). A lie algebra is a vector space g over a field $\mathcal{F}$ equipped with a vector product called the Lie bracket $[\cdot, \cdot] : g \times g \rightarrow g$. The Lie bracket satisfies the following properties:

i) Bilinearity

$$[ax + by, z] = a[x, z] + b[y, z] \quad (115a)$$

$$[z, ax + by] = a[z, x] + b[z, y], \quad (115b)$$

ii) Alternativity

$$[x, x] = 0, \quad (116)$$

iii) Jacobi Identity

$$[[x, y], z] + [[y, z], x] + [[z, x], y] = 0. \quad (117)$$

Bilinearity and Alternativity imply Anti-commutativity $[x, y] = -[y, x]$.

The simple compact Lie groups can be neatly classified into 4 classical Lie groups and 5 exceptional Lie groups given by the following Dynkin diagrams.

A_n, B_n, C_n , and D_n refer to the groups of $SU(n+1)$, $SO(2n+1)$, $Sp(2n)$, and $SO(2n)$ respectively. G_2, F_4, E_6, E_7 , and E_8 are five exceptional Lie groups.

The classical Lie algebras can be expressed in terms of real numbers $\mathbb{R}$, complex numbers $\mathbb{C}$, and quaternions $\mathbb{H}$.

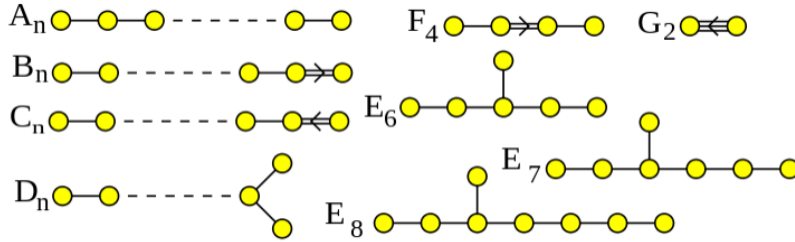


Figure 7: Dynkin diagrams for Simple Compact Lie Algebras (Credits: Wikipedia)

$$\mathfrak{so}(n) = \{x \in \mathbb{R}[n], \quad x^\dagger = -x, \quad \text{tr}(x) = 0\} \quad (118a)$$

$$\mathfrak{su}(n) = \{x \in \mathbb{C}[n], \quad x^\dagger = -x, \quad \text{tr}(x) = 0\} \quad (118b)$$

$$\mathfrak{sp}(n) = \{x \in \mathbb{H}[n], \quad x^\dagger = -x, \quad \text{tr}(x) = 0\} \quad (118c)$$

All the exceptional Lie algebras are related to the octonions $\mathbb{O}$ and can be understood via the magic square [22].

In 1956, Rosenfeld proposed that F_4 might be the isometry group of $\mathbb{OP}^2$. The exceptional Lie groups E_6 , E_7 , and E_8 are also related to $(\mathbb{C} \otimes \mathbb{O})\mathbb{P}^2$, $(\mathbb{H} \otimes \mathbb{O})\mathbb{P}^2$, and $(\mathbb{O} \otimes \mathbb{O})\mathbb{P}^2$ respectively. In around 1958 Tits [19] and Freudenthal [18] came up with a construction called the magic square of Lie algebras that nicely decomposes certain Lie algebras in terms of sub-algebras corresponding to two different division algebras. The magic square we present here was constructed by Vinberg [20] in 1966. Corresponding to two different division algebras $\mathbb{K}_1, \mathbb{K}_2$ there exist a Lie algebra given by:

$$L[\mathbb{K}_1, \mathbb{K}_2] = A'_3(\mathbb{K}_1 \otimes \mathbb{K}_2) + \text{Der}(\mathbb{K}_1) + \text{Der}(\mathbb{K}_2). \quad (119)$$

Where, $\text{Der}(\mathbb{K})$ is the derivation algebra of $\mathbb{K}$ and $A'_3(\mathbb{K}_1 \otimes \mathbb{K}_2)$ is a 3×3 anti-hermitian traceless matrix. Using this construction we get the following table:

As can be seen from the table 8, all the exceptional Lie algebras:- $\mathfrak{g}_2, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7$, and $\mathfrak{e}_8$ are related to the octonions.

Jordan Algebras

In 1932, Pascual Jordan came up with the notion of Jordan algebra to propose a commutative but non-associative algebra of observables in quantum

	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	$\mathbb{O}$
$\mathbb{R}$	$\mathfrak{so}(3)$	$\mathfrak{su}(3)$	$\mathfrak{sp}(3)$	$\mathfrak{f}_4$
$\mathbb{C}$	$\mathfrak{su}(3)$	$\mathfrak{su}(3) \oplus \mathfrak{su}(3)$	$\mathfrak{su}(6)$	$\mathfrak{e}_6$
$\mathbb{H}$	$\mathfrak{sp}(3)$	$\mathfrak{su}(6)$	$\mathfrak{so}(12)$	$\mathfrak{e}_7$
$\mathbb{O}$	$\mathfrak{f}_4$	$\mathfrak{e}_6$	$\mathfrak{e}_7$	$\mathfrak{e}_8$

Figure 8: The Magic Square of Lie Algebras.

mechanics [48]. A Jordan algebra is a vector space $\mathcal{J}$ over scalar fields $\mathcal{F}$ equipped with a bi-product called the Jordan product $\circ : \mathcal{J} \times \mathcal{J} \rightarrow \mathcal{J}$ given by:

$$x \circ y = \frac{1}{2}(xy + yx). \quad (120)$$

A real Jordan algebra or a Euclidean Jordan algebra is a Jordan algebra that satisfies the following property:

$$a_1^2 + a_2^2 + \dots + a_n^2 = 0 \Rightarrow a_i = 0 \quad \forall i. \quad (121)$$

Real Jordan algebras are naturally power associative and therefore satisfy the following identity called the Jordan identity:

$$(x \circ y) \circ x^2 = x \circ (y \circ x^2). \quad (122)$$

Analogously to simple compact Lie algebras, simple real Jordan algebras can be classified into 4 classical Jordan algebras (depend on $\mathbb{R}, \mathbb{C}, \mathbb{H}$) and 1 exceptional Jordan algebra (depends on $\mathbb{O}$).

If we define $H_n[\mathbb{K}]$ to be $n \times n$ hermitian matrices with entries in $\mathbb{K}$ then the simple real Jordan algebras can be classified as:

$$\begin{aligned}
H_n[\mathbb{R}] & \text{ with the product } a \circ b = \frac{1}{2}(ab + ba) \\
H_n[\mathbb{C}] & \text{ with the product } a \circ b = \frac{1}{2}(ab + ba) \\
H_n[\mathbb{H}] & \text{ with the product } a \circ b = \frac{1}{2}(ab + ba) \\
\mathbb{R}^n \oplus \mathbb{R} & \text{ with the product } (\alpha, a) \circ (\beta, b) = (a\beta + b\alpha, \alpha \cdot \beta + ab) \\
H_3[\mathbb{O}] & \text{ with the product } a \circ b = \frac{1}{2}(ab + ba)
\end{aligned} \quad (123)$$

B Real Slices of Twistor Space

- Take a real group $\mathcal{G} \subset \text{SO}(6, \mathbb{C})$.
- Restrict twistors to be spinors of $\mathcal{G}$.
- This imposes conditions on ω and π .
- Use the incidence relation to obtain restrictions on spacetime.

If $\mathcal{G}$ is a unitary group then there will be a natural group invariant inner product between ω and π . Note that the existence of spacetime with a real signature is not guaranteed.

Example: $\text{SO}(3, 3)$

$\text{SO}(3, 3)$ is another real subgroup of $\text{SO}(6, \mathbb{C})$. The double cover of $\text{SO}(3, 3)$ is $\text{Sl}(4, \mathbb{R})$. Thus, if the twistors are spinors of $\text{SO}(3, 3)$ they must be real.

$$\Rightarrow \omega^A, \pi_{\dot{A}} \in \mathbb{R} \quad (124)$$

But $\omega^A = ix^{A\dot{A}}\pi_{\dot{A}}$, therefore spacetime points will be identified by 2×2 real matrices ($\mathbb{R}[2]$).

$$\therefore ix = \frac{1}{\sqrt{2}} \begin{pmatrix} x_0 + x_3 & x_1 - x_2 \\ x_1 + x_2 & x_0 - x_3 \end{pmatrix} \in \mathbb{R}[2] \quad (125a)$$

$$\Rightarrow x = \frac{-i}{\sqrt{2}} \begin{pmatrix} x_0 + x_3 & x_1 - x_2 \\ x_1 + x_2 & x_0 - x_3 \end{pmatrix} \quad (125b)$$

$$\Rightarrow \det(x) = x_3^2 + x_1^2 - x_0^2 - x_2^2 \quad (125c)$$

Therefore the spacetime point we get from the incidence relation has a (2,2) split signature. The isometry group in such a spacetime is $\text{SO}(2, 2)$ and the conformal group is $\text{SO}(3, 3)$. Therefore, our result is consistent.

Example: $\text{SO}(1, 5)$

$\text{SO}(1, 5)$ is yet another real subgroup of $\text{SO}(6, \mathbb{C})$. The double cover of $\text{SO}(1, 5)$ is $\text{Sl}(2, \mathbb{H})$. Therefore, in this case, the twistor space $\mathbb{T} \cong \mathbb{C}^4$ is

identified with $\mathbb{H}^2$. The Grassmanian $G(2, \mathbb{C}^4)$ that corresponds to spacetime is identified with $\mathbb{H}$ and the sections of the tautological spinor bundle also live in $\mathbb{H}$. Therefore in this case a spacetime point can be written as:

$$ix = x_0 + x_1\hat{i} + x_2\hat{j} + x_3\hat{k} \in \mathbb{H}. \quad (126)$$

Invoking the isomorphism between quaternions and Pauli matrices we have:

$$ix = x_0\mathbb{I} - i(x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3) = \begin{pmatrix} x_0 - ix_3 & -ix_1 + x_2 \\ -ix_1 - x_2 & x_0 + ix_3 \end{pmatrix}. \quad (127)$$

Therefore we have:

$$\det(x) = -x_0^2 - x_1^2 - x_2^2 - x_3^2. \quad (128)$$

Therefore the spacetime we get in this case has $(0, 4)$ Euclidean signature with the isometry group $\text{SO}(0, 4)$. The conformal group of $\text{SO}(0, 4)$ is $\text{SO}(1, 5)$, this is a consistency check.

Note that all real slices do not need to correspond to spacetime with a real signature. Let's see the example of the subgroup $\text{SO}(6) \subset \text{SO}(6, \mathbb{C})$ that does not correspond to any real spacetime.

Example: $\text{SO}(6)$

$\text{SO}(6)$ is another real subgroup of $\text{SO}(6, \mathbb{C})$. The double cover of $\text{SO}(6)$ is $\text{SU}(4)$. This is a unitary group and has an invariant product on twistors given by:

$$Z^\dagger Z. \quad (129)$$

We have:

$$(Z^\alpha)^\dagger Z^\alpha = \omega^\dagger \omega + \pi^\dagger \pi \quad (130)$$

$$= \pi^\dagger (x^\dagger x + \mathbb{I}) \pi. \quad (131)$$

Demanding that this inner product be $\text{SU}(4)$ invariant does not give us a real spacetime since conditions on $x^\dagger x$ will give us a quadric rather than a vector space.

C Quantized Twistors and Conformal Algebra

Define the four-momentum as:

$$p_a = \bar{\pi}_A \pi_{\dot{A}} \quad (132)$$

The angular momentum is given as:

$$M_{ab} = x_a p_b - x_b p_a \quad (133)$$

$$= x_{A\dot{A}} \bar{\pi}_B \pi_{\dot{B}} - x_{B\dot{B}} \bar{\pi}_A \pi_{\dot{A}} \quad (134)$$

$$= \bar{\pi}_B x_{A\dot{A}} \pi_{\dot{B}} - \bar{\pi}_A x_{B\dot{B}} \pi_{\dot{A}} \quad (135)$$

$$= 2(\bar{\pi}_{(B} x_{A)[\dot{A}} \pi_{\dot{B}]} - \bar{\pi}_{[A} x_{B]\dot{B}} \pi_{\dot{A}}) \quad (136)$$

Using the twistor incidence relation, this simplifies to:

$$M_{ab} = i\bar{\pi}_{(A} \omega_{B)} \epsilon_{\dot{A}\dot{B}} - i\epsilon_{AB} \bar{\omega}_{(\dot{A}} \pi_{\dot{B})} \quad (137)$$

Twistor commutation relations can give the conformal algebra. Let us consider the following commutation relations:

$$[Z^\alpha, Z^\beta] = 0 \quad (138a)$$

$$[\bar{Z}_\alpha, \bar{Z}_\beta] = 0 \quad (138b)$$

$$[Z^\alpha, \bar{Z}_\beta] = \delta_\beta^\alpha \quad (138c)$$

If we expand the twistor Z^α and its dual twistor $\bar{Z}_\alpha$ as

$$Z^\alpha = (\omega^A, \pi_{\dot{A}}) \quad (139a)$$

$$\bar{Z}_\alpha = (\bar{\pi}_A, \bar{\omega}^{\dot{A}}) \quad (139b)$$

then Eq. (138a) becomes

$$[\omega^A, \omega^B] = 0 \quad (140a)$$

$$[\pi_{\dot{A}}, \pi_{\dot{B}}] = 0 \quad (140b)$$

$$[\omega^A, \pi_{\dot{B}}] = 0 \quad (140c)$$

$$[\pi_{\dot{A}}, \omega^B] = 0 \quad (140d)$$

Eq. (138b) becomes

$$[\bar{\pi}_A, \bar{\pi}_B] = 0 \quad (141a)$$

$$[\bar{\omega}^{\dot{A}}, \bar{\omega}^{\dot{B}}] = 0 \quad (141b)$$

$$[\bar{\pi}_A, \bar{\omega}^{\dot{B}}] = 0 \quad (141c)$$

$$[\bar{\omega}^{\dot{A}}, \bar{\pi}_B] = 0 \quad (141d)$$

and Eq. (138c) becomes

$$[\omega^A, \bar{\pi}_B] = \delta_B^A \quad (142a)$$

$$[\pi_{\dot{A}}, \bar{\omega}^{\dot{B}}] = \delta_{\dot{A}}^{\dot{B}} \quad (142b)$$

$$[\omega^A, \bar{\omega}^{\dot{B}}] = 0 \quad (142c)$$

$$[\pi_{\dot{A}}, \bar{\pi}_B] = 0 \quad (142d)$$

or, in other words, the commutator of two unbarred or two barred quantities vanish, while (with indices lowered) the commutator of an unbarred and a barred quantity gives

$$[\omega_A, \bar{\pi}_B] = \epsilon_{BA} \quad (143a)$$

$$[\pi_{\dot{A}}, \bar{\omega}_{\dot{B}}] = \epsilon_{\dot{A}\dot{B}} \quad (143b)$$

$$[\omega_A, \bar{\omega}_{\dot{B}}] = 0 \quad (143c)$$

$$[\pi_{\dot{A}}, \bar{\pi}_B] = 0. \quad (143d)$$

Now we want to show that the above commutation relations naturally give us the Poincare algebra.

$$[p_a, p_b] = 0 \quad (144a)$$

$$[M_{ab}, p_c] = i(p_a \eta_{bc} - p_b \eta_{ac}) \quad (144b)$$

$$[M_{ab}, M_{cd}] = i(\eta_{ad} M_{bc} + \eta_{bc} M_{ad} - \eta_{ac} M_{bd} - \eta_{bd} M_{ac}) \quad (144c)$$

Proving 144a is trivial. 144b simplifies to:

$$[M_{ab}, p_c] = i(\epsilon_{AB} \bar{\pi}_C \epsilon_{\dot{C}}(\dot{A} \pi_{\dot{B}}) - \bar{\pi}_{(A} \epsilon_{B)C} \pi_{\dot{C}} \epsilon_{\dot{A}\dot{B}}) \quad (145)$$

To check that 145 agrees with 144b we need to use $\eta_{ab} = \epsilon_{AB}\epsilon_{\dot{A}\dot{B}}$, and $T_{C[AB]} = \frac{1}{2}T_{CD}{}^D\epsilon_{AB}$ to find that:

$$[M_{ab}, p_c] = i(p_a\eta_{bc} - p_b\eta_{ac}) \quad (146a)$$

$$= -i(\bar{\pi}_A\epsilon_{BC}\epsilon_{\dot{C}\dot{B}}\pi_{\dot{A}} - \bar{\pi}_B\epsilon_{AC}\epsilon_{\dot{C}\dot{A}}\pi_{\dot{B}}) \quad (146b)$$

$$= -2i(\bar{\pi}_{(A}\epsilon_{B)C}\epsilon_{\dot{C}[\dot{B}}\pi_{\dot{A}]} - \bar{\pi}_{[B}\epsilon_{A]C}\epsilon_{\dot{C}(\dot{A}}\pi_{\dot{B}]}) \quad (146c)$$

$$= -i(\bar{\pi}_{(A}\epsilon_{B)C}\epsilon_{\dot{C}\dot{D}}\pi^{\dot{D}}\epsilon_{\dot{B}\dot{A}} - \epsilon_{BA}\bar{\pi}_D\epsilon^D{}_{\dot{C}}\epsilon_{\dot{A}}\pi_{\dot{B}}) \quad (146d)$$

$$= -i(\bar{\pi}_{(A}\epsilon_{B)C}\pi_{\dot{C}}\epsilon_{\dot{A}\dot{B}} - \epsilon_{AB}\bar{\pi}_C\epsilon_{\dot{C}(\dot{A}}\pi_{\dot{B}})) \quad (146e)$$

$$= i(\epsilon_{AB}\bar{\pi}_C\epsilon_{\dot{C}(\dot{A}}\pi_{\dot{B}}) - \bar{\pi}_{(A}\epsilon_{B)C}\pi_{\dot{C}}\epsilon_{\dot{A}\dot{B}}). \quad (146f)$$

Thus, 144b follows naturally from twistor commutation relations. To check 144c, we obtain the following expression:

$$[M_{ab}, M_{cd}] = \frac{1}{2}\epsilon_{\dot{A}\dot{B}}\epsilon_{\dot{C}\dot{D}}(\epsilon_{AC}\bar{\pi}_{(B}\omega_{D)} + \epsilon_{BD}\bar{\pi}_{(A}\omega_{C)}) \quad (147)$$

To check that (147) agrees with (144c), start by rewriting (144c) as

$$[M_{ab}, M_{cd}] = -4i\eta_{[a[c}M_{b]d]} \quad (148)$$

which is anti-symmetrized over the pair a and b , and also over c and d . If we then apply identity (3.4.55) in [25]:

$$H_{[ab]} = H_{[AB](\dot{A}\dot{B})} + H_{(AB)[\dot{A}\dot{B}]} = \frac{1}{2}\epsilon_{AB}H_C{}^C{}_{(\dot{A}\dot{B})} + \frac{1}{2}\epsilon_{\dot{A}\dot{B}}H_{(AB)\dot{C}}{}^{\dot{C}} \quad (149)$$

to each of these pairs, we find that (148) precisely reduces to (147).

Above we have shown that Poincare algebra follows from quantized twistors. One can further show that conformal algebra follows in a similar way. The Dilation operator is given by:

$$D = \frac{-i}{2}(\pi_A\bar{\omega}^A - \omega_{\dot{A}}\bar{\pi}^{\dot{A}}) \quad (150)$$

and the special conformal transformations are generated by:

$$K_{A\dot{A}} = \omega_{\dot{A}}\bar{\omega}_A. \quad (151)$$

D Quantized Supertwistors and Super Conformal Algebra

Quantized supertwistors follow the following commutation relations:

$$[\omega^A, \bar{\pi}_B] = \delta_B^A \quad (152a)$$

$$[\pi_{\dot{A}}, \bar{\omega}^{\dot{B}}] = \delta_{\dot{A}}^{\dot{B}} \quad (152b)$$

$$[\omega^A, \bar{\omega}^{\dot{B}}] = 0 \quad (152c)$$

$$[\pi_{\dot{A}}, \bar{\pi}_B] = 0. \quad (152d)$$

With lowered indices:

$$[\omega_A, \bar{\pi}_B] = \epsilon_{BA} \quad (153a)$$

$$[\pi_{\dot{A}}, \bar{\omega}_{\dot{B}}] = \epsilon_{\dot{A}\dot{B}} \quad (153b)$$

$$[\omega_A, \bar{\omega}_{\dot{B}}] = 0 \quad (153c)$$

$$[\pi_{\dot{A}}, \bar{\pi}_B] = 0. \quad (153d)$$

The fermionic commutation rules are:

$$\{\Psi^i, \bar{\Psi}_j\}_+ = \delta_j^i \quad (154a)$$

$$\{\Psi^i, \Psi^j\}_+ = 0 \quad (154b)$$

$$\{\bar{\Psi}_i, \bar{\Psi}_j\}_+ = 0. \quad (154c)$$

All the bosonic twistor variables commute with fermionic variables.

The supercharges and the conformal supercharges are given by:

$$\bar{Q}^{\dot{A}i} = -i\bar{\pi}^{\dot{A}}\Psi^i, \quad Q_i^A = i\pi^A\bar{\Psi}_i. \quad (155a)$$

$$\bar{S}_{\dot{A}i} = i\omega_{\dot{A}}\bar{\Psi}_i, \quad S_i^A = -i\bar{\omega}_A\Psi_i. \quad (155b)$$

We have already shown that conformal algebra can be obtained from quantized twistors in Appendix B. Using the (anti-)commutation rules given here, it is easy to show that:

$$\{Q_A^i, Q_B^j\} = \{\bar{Q}_{\dot{A}}^i, \bar{Q}_{\dot{B}}^j\} = 0 \quad (156a)$$

$$\{S, S\} = \{\bar{S}, \bar{S}\} = 0 \quad (156b)$$

$$\{Q, \bar{S}\} = \{\bar{Q}, S\} = 0 \quad (156c)$$

$$[Q, p] = [\bar{Q}, p] = 0 \quad (156d)$$

$$[S, k] = [\bar{S}, k] = 0 \quad (156e)$$

Let us show that:

$$[D, Q] = \frac{i}{2}Q \quad (157)$$

$$\begin{aligned} [D, Q_B^i] &= \frac{-i}{2} [\pi_A \bar{\omega}^A - \omega_A \bar{\pi}^{\dot{A}}, i\pi_B \bar{\Psi}^i] \\ &= \frac{\pi_A}{2} [\bar{\omega}^A, \pi_B] \bar{\Psi}^i \\ &= \frac{-\pi_A}{2} \delta_B^A \bar{\Psi}^i \\ &= \frac{i}{2} Q_B^i. \end{aligned} \quad (158)$$

Similarly, it can be shown that:

$$[D, Q] = \frac{i}{2}Q, \quad [D, \bar{Q}] = \frac{i}{2}\bar{Q} \quad (159a)$$

$$[D, S] = \frac{-i}{2}S, \quad [D, \bar{S}] = \frac{-i}{2}\bar{S} \quad (159b)$$

Also:

$$\begin{aligned} \{Q_A^i, \bar{Q}_{\dot{B}}^j\} &= \pi_A \bar{\Psi}^i \bar{\pi}_{\dot{B}} \Psi^j + \bar{\pi}_{\dot{B}} \Psi^j \pi_A \bar{\Psi}^i \\ &= \pi_A \bar{\pi}_{\dot{B}} \{\bar{\Psi}^i, \Psi^j\} \\ &= \delta^{ij} \pi_A \bar{\pi}_{\dot{B}} \\ &= 2\delta^{ij} (\sigma^\mu)_{A\dot{B}} p_\mu \end{aligned} \quad (160)$$

Similarly, we can prove the following:

$$\{S_A^i, \bar{S}_{\dot{B}}^j\} = 2\delta^{ij} (\sigma^\mu)_{A\dot{B}} k_\mu. \quad (161)$$

E Quantized Supertwistors and Super(Split-)Octonions

A quantized supertwistor algebra is given by:

$$\mathcal{A} = A^\Omega \hat{\bar{Z}}_\Omega + \bar{A}_\Omega \hat{Z}^\Omega \quad (162a)$$

$$= A^\alpha \bar{Z}_\alpha + \bar{A}_\alpha Z^\alpha + A^i \bar{\psi}_i + \bar{A}_i \psi^i \in \mathcal{O}, \quad (162b)$$

where, both A^α and A^i are commuting c-valued coefficients. The inner-product $A^\alpha \bar{Z}_\alpha$ is $\mathfrak{su}(2, 2)$ invariant whereas, the inner-product $A^i \bar{\Psi}_i$ is $\mathfrak{su}(4)$ invariant. The supertwistor commutation rules are given by:

$$[\hat{Z}^\Omega, \bar{\hat{Z}}_\Theta]_\mp = \delta_\Theta^\Omega, \quad [\hat{Z}^\Omega, \hat{Z}^\Theta]_\mp = 0, \quad [\bar{\hat{Z}}_\Omega, \bar{\hat{Z}}_\Theta]_\mp = 0. \quad (163)$$

More elaborately they can be written as:

$$[Z^\alpha, \bar{Z}_\beta] = \delta_\beta^\alpha, \quad [Z^\alpha, Z_\beta] = [\bar{Z}^\alpha, \bar{Z}_\beta] = 0 \quad (164a)$$

$$\{\Psi^i, \bar{\Psi}_j\} = \delta_j^i, \quad \{\Psi^i, \Psi_j\} = \{\bar{\Psi}^i, \bar{\Psi}_j\} = 0 \quad (164b)$$

$$[Z^\alpha, \Psi^i] = [Z^\alpha, \bar{\Psi}^i] = [\bar{Z}^\alpha, \Psi^i] = 0. \quad (164c)$$

Introduce the following map:

$$\mathbf{I}_1(\mathcal{B}) := i(\bar{B}_\alpha Z^\alpha + B^\alpha \bar{Z}_\alpha + B^i \bar{\psi}_i - \bar{B}_i \psi^i). \quad (165)$$

Let us define the following triple product:

$$[\mathcal{A}, \mathcal{B}, \mathcal{C}] := \sum_{\odot} \mathcal{A}[\mathcal{B}, (\mathbf{I}_1 \mathcal{C})]_\mp \quad (166a)$$

$$= i \sum_{\odot} \mathcal{A}(\bar{B}_\alpha C^\alpha - B^\alpha \bar{C}_\alpha + B^i \bar{C}_i - \bar{B}_i C^i) \in \mathcal{O}. \quad (166b)$$

Eqn.(166b) follows from eqn.(166a) using the supertwistor (anti-)commutation rules. Here $[\mathcal{B}, (\mathbf{I}_1 \mathcal{C})]_\mp$ is defined in the following way. Whenever there is a bosonic contribution $(Z^\alpha, \bar{Z}_\alpha)$ from either $\mathcal{B}$ or $\mathcal{C}$ there is a negative sign between whereas when there is a purely fermionic contribution $(\Psi^i, \bar{\Psi}_i)$ from $\mathcal{B}$ and $\mathcal{C}$ there is a positive sign. This is simply the supercommutator.

Now we can choose a purely bosonic identity element in the algebra:

$$\mathcal{E} = \bar{E}_\alpha Z^\alpha + E_\alpha \bar{Z}_\alpha, \quad (167)$$

which breaks the symmetry from $\mathfrak{su}(2, 2)$ to $\mathfrak{g}_2 \oplus \mathfrak{g}_2^*$. Any element of the algebra can be broken into components parallel and perpendicular to $\mathcal{E}$. The bilinear product in the algebra is simply the triple product of any two elements in the algebra along with the identity.

The inner product can be defined in the following way:

$$\mathcal{A} \cdot \mathcal{B} = i[\mathcal{A}, (\mathbf{I}_2 \mathcal{B})]_\mp \quad (168a)$$

$$= (\bar{A}_\alpha B^\alpha + A^\alpha \bar{B}_\alpha - A^i \bar{B}_i - \bar{A}_i B^i). \quad (168b)$$

Eqn.(168b) follows from eqn.(168a) using the supertwistor (anti-)commutation rules. ‘ $\mathbf{I}_2$ ’ is defined as follows:

$$\mathbf{I}_2(\mathcal{B}) := i(\bar{B}_\alpha Z^\alpha - B^\alpha \bar{Z}_\alpha + B^i \bar{\psi}_i + \bar{B}_i \psi^i). \quad (169)$$

Note that the first part of the inner product given by $\bar{A}_\alpha B^\alpha + A^\alpha \bar{B}_\alpha$ has a (4, 4) split-signature whereas the second part of the inner product given by $-A^i \bar{B}_i - \bar{A}_i B^i$ has an (8, 0) definite signature. The algebra obtained in this case is that of $\tilde{\mathbb{O}} \oplus \mathbb{O}$.

The algebra $\mathcal{O} \cong \tilde{\mathbb{O}} \oplus \mathbb{O}$ is a superalgebra that naturally splits as a direct sum of a bosonic part $A^\alpha \bar{Z}_\alpha + \bar{A}_\alpha Z^\alpha \in \tilde{\mathbb{O}}$ and a fermionic part $A^i \bar{\psi}_i + \bar{A}_i \psi^i \in \mathbb{O}$.

Freed and Moore talk about ‘the tenfold way’, which is a classification of all the associative superdivision algebras in [49]. The original credit for this goes back to Freeman Dyson [50]. The algebra $\tilde{\mathbb{O}} \oplus \mathbb{O}$, however, is not associative, nor is it a division algebra due to the presence of split-octonions but it is a supercomposition algebra.